

# MetroMeal Delivery: Can you trace a business story from decision through KPI, grain, chart, claim, and validation?

Your guide for the lab. Start with the email, then work each card in order. Everything else is optional reference.

## Contents

The request

At a glance

Before class

Part A: What decision can the KPI inform?

Part B: Is MetroMeal's latest week below target?

Part C: Is the KPI calculation reproducible?

Part D: What is the four-week on-time rate, built from counts?

Part E: What does one row represent?

Part F: How many orders hide in four joined rows?

Part G: Does the chart communicate honestly?

Part H: What makes the MetroMeal dashboard chart misleading?

Part I: Is the data story traceable and bounded?

Part J: Can every number in the MetroMeal story be traced?

Part K: What does one row of the MetroMeal file represent?

Part L: How did each week compare with the target and the guardrail?

Part M: Does the chart show the checked numbers honestly?

Part N: What should the director do next?

Finish

Checklist

Optional practice

## The request

Subject: **Weekend staffing: a checked KPI before the next scheduling cycle**



**Regional Operations Director**

To: Regional Operations Analyst (you)

Cc: the scheduling team

Hi Operations Analytics team,

Before I set the next staffing cycle, I need a checked view of our recent delivery performance. The latest dashboard looks below target, and weekend staffing has already been suggested as the reason. Warm up on five short practice cases (other organizations, made-up numbers), each followed by the same check on our MetroMeal figures. Then build the real KPI from our delivery file.

Specifically, I need answers to 14 questions:

**1. Practice: what decision can HarborCare's KPI inform?**

Frame the clinic manager's decision and check the no-show rate against its target and guardrail.

Part A

**2. Is our latest week below target?**

Use the slide numbers for the latest week. Tell me the on-time rate, the gap in percentage points, and the complaint rate.

Part B

**3. Practice: is NorthLine's KPI reproducible?**

Could another analyst get the same number from your definition alone?

Part C

**4. What is our four-week rate?**

Build it from counts, not from an average of weekly percentages.

Part D

**5. Practice: what does one BrightCart row represent?**

Classify the fields and warn me about the order-to-item join.

Part E

**6. How many orders hide in our joined rows?**

Show me the difference between joined rows and distinct orders.

Part F

**7. Practice: is RiverTown's chart honest?**

Name what is misleading and how you would fix it.

Part G

**8. What is wrong with our dashboard chart?**

It says performance *collapses*. Check the scale, the colors and the title.

Part H

**9. Practice: is GreenBlock's story bounded?**

Separate what the records show from what the draft claims.

Part I

**10. Can every number in our story be traced?**

Verify at least one number a second way before it goes in the story.

Part J

**11. What does one row of our delivery file represent?**

Part K

Confirm the key, the grain and which deliveries are eligible.

**12. How did each week compare with the target and the complaint guardrail?**

Part L

Build the weekly table from the raw rows and check the late deliveries in the latest week.

**13. Can you give me one honest chart?**

Part M

Linked to your checked numbers, with a zero baseline, labels, denominators and the target line.

**14. Does the evidence support investigating weekend staffing?**

Part N

Three sentences: what it shows, what it does not show, and what I should do next. Do not claim a cause the records cannot show.

Please send me:

- your saved practice workbook (nothing to upload; this is ungraded practice), and
- three bounded sentences in **Response!B7** that the instructor may read aloud for feedback.

Our delivery file and the practice workbook are attached. Thanks. I want an answer I can defend, not a guess.

Best,

**Regional Operations Director**

MetroMeal Delivery



orders\_story.csv



S02P\_Learning\_Workbook.xlsx



S02P\_SPSS.sps

## Your lab at a glance

**Before class**

setup  
untimed

**Part A**

3 min  
Slide 19 → 20

**Part B**

3 min  
Slide 19 → 20

**Part C**

3 min  
Slide 38 → 39

**Part D**

3 min  
Slide 38 → 39

**Part E**

3 min  
Slide 54 → 55

**Part F**

3 min  
Slide 54 → 55

**Part G**

3 min  
Slide 70 → 71

**Part H**

3 min  
Slide 70 → 71

**Part I**

3 min  
Slide 78 → 79

**Part J**

3 min  
Slide 78 → 79

**Part K**

4 min  
In class

**Part L**

4 min  
In class

**Part M**

4 min  
In class

**Part N**

3 min  
In class

**Finish**

checklist  
checklist

## Before class: get ready (untimed)

- 1** Unzip `S02P_Student_Package.zip` into one folder.  
Keep every file together. SPSS looks for `orders_story.csv` next to the syntax.
- 2** Open `S02P_Learning_Workbook.xlsx`, read **Start\_Here**, then save a copy with your own name.  
Pattern: `S02P_FirstName_LastName_NetID.xlsx` (NetID is the part before @nyu.edu).
- 3** Open `S02P_SPSS.sps` in SPSS and fix line 7.  
Replace `CHANGE_TO_YOUR_EXTRACTED_LAB_FOLDER` with your folder's path.
- 4** Select the first block (Import `orders_story.csv`) and choose **Run > Selection**.

✓ **Ready when:** SPSS reports 2,000 eligible completed deliveries and your workbook is saved under your own name.

⚠ **Stuck?** Ask for help before the first practice. This session is ungraded practice.

## Find your folder path (for the `CD` line)

SPSS needs the path of the **folder** you unzipped, the one that holds `S02P_SPSS.sps` and the CSV files. It is **not** the path to the `.sps` file. If you open the folder and see the `.sps` file and the CSV files inside, that is the right folder.

### Windows

1. Unzip first: right-click `S02P_Student_Package.zip` > **Extract All** > **Extract**. Do not work inside the zip.
2. Open the extracted `S02P_Student_Package` folder, then the **Student** folder inside it. You should now see `S02P_SPSS.sps` and the CSV files. This is your lab folder.
3. Click once in the address bar at the top of File Explorer: it turns into the path. Press **Ctrl+C**.
4. Or: go up one level, hold **Shift**, right-click that folder > **Copy as path**, then delete the double quotes it adds.

Looks like

```
C:\Users\you\Downloads\S02P_Student_Package\Student
```

### Mac

1. Unzip first: double-click `S02P_Student_Package.zip` in Finder.
2. Open the extracted `S02P_Student_Package` folder, then the **Student** folder inside it. You should now see `S02P_SPSS.sps` and the CSV files. This is your lab folder.
3. Go up one level and click that folder **once** to select it (the folder, not a file inside it). Press **Option+Command+C** (Copy as Pathname), or right-click it, hold **Option** and choose **Copy ... as Pathname**.

Looks like

```
/Users/you/Downloads/S02P_Student_Package/Student
```

Paste it between the single quotes on the `CD` line:

```
CD 'C:\Users\you\Downloads\S02P_Student_Package\Student'.  
CD '/Users/you/Downloads/S02P_Student_Package/Student'.
```

\* *Windows*

\* *Mac*

⚠ **Common mistakes:** the path ends in `/S02P_SPSS.sps` (remove the file name); the quotes or the final period are missing; the files are still inside the zip; OneDrive or iCloud has not finished downloading the folder. If H0 says a file cannot be found, check the path first. If the path itself contains an apostrophe ( `'` ), put the path in double quotes instead, for example `CD "/Users/you/Kim's files/S02P_Student_Package/Student"`.

## How every card works

Yellow cells: you type here    Lavender cells: guidance

Each card opens with a short recap of the session content, with the slides to reread. Type each formula starting with `=`, then press **Enter**. Some gaps are in percentage points, so a few formulas multiply by 100 on purpose: follow the card exactly.

## Part A · What decision can the KPI inform?

[From slide 19](#) → work this card → [Back to slide 20](#)

· bring: One bounded decision question, one supported descriptive claim, and one causal claim to remove.

### WHAT THE SESSION COVERED · SECTION 1 DECISION FRAME AND RUNNING-CASE BRIEF

Start with the **decision**, not the chart. Name who acts, what they might do, and by when. Then name the measures:

- **KPI:** the measure you monitor. Here: the monthly no-show rate.  $\text{rate} = \frac{\text{qualifying count}}{\text{eligible count}}$
- **Target:** the benchmark for the KPI. HarborCare wants no more than 10.0% no-shows.
- **Guardrail:** a limit that protects a *different* outcome. Complaints must stay at or below 4.0%.
- **Evidence boundary:** one month of counts can show a level. It cannot show a trend or a cause.

HarborCare is a made-up clinic. Its numbers are not part of the MetroMeal data.

#### COURSE SLIDES

Slides 9–18  
Decision-First Analysis 10–11 · KPI,  
Target, and Guardrail 12–14 ·  
MetroMeal's Bounded Question 15–  
16 · Retrieval Check 18

#### CONCISE TOOLKIT

pp. 4–7  
Decision-first analysis p. 4 · KPI,  
target and guardrail p. 7

#### FULL TOOLKIT

pp. 4–17  
Decision-first analysis p. 9

### THE EQUATIONS YOU ARE BUILDING

$N$  = no-shows ·  $C$  = complaints ·  $E$  = eligible appointments

$$\text{No-show rate} = \frac{N}{E} = \frac{B12}{B13} \quad \text{Complaint rate} = \frac{C}{E} = \frac{B14}{B13}$$

### OPEN

Open `Class_Practice!B6`.

### FILL IN THE YELLOW CELLS

| # | Cell    | Type this                                        | In math                      | Check       |
|---|---------|--------------------------------------------------|------------------------------|-------------|
| 1 | B10:B11 | <code>0.10</code> <code>0.04</code>              | $T, G$ max target, guardrail | 10.0%, 4.0% |
| 2 | B12:B14 | <code>46</code> <code>400</code> <code>12</code> | $N, E, C$ counts             |             |
| 3 | B15     | <code>=B12/B13</code>                            | $N/E$ no-show rate           | 11.5%       |
| 4 | B16     | <code>=B14/B13</code>                            | $C/E$ complaint rate         | 3.0%        |

| # | Cell | Type this                                                                                                                       | In math | Check |
|---|------|---------------------------------------------------------------------------------------------------------------------------------|---------|-------|
|   |      | <b>B6:</b> Who acts? (the HarborCare clinic manager)                                                                            |         |       |
|   | B6   | <b>B7:</b> What might they do? Keep it to investigating or revising reminder scheduling.                                        |         |       |
|   | B7   | <b>B8:</b> By when? (before the next reminder-scheduling decision)                                                              |         |       |
| 5 | B8   | <b>B9:</b> Name the KPI with its population, e.g. the monthly no-show rate among eligible appointments.                         |         |       |
|   | B9   |                                                                                                                                 |         |       |
|   | B17  | <b>B17:</b> One supported claim; one causal or trend claim to remove; then a revised question that asks whether to investigate. |         |       |

✓ **Done when:** B15 shows 11.5% and B16 shows 3.0%. B17 has a supported claim, a claim to remove and a revised question. **Save.**

⚠ **Most common mistake:** treating the complaint result as good news about no-shows. The two rates answer different questions. One can pass while the other fails.

no-shows:  $\frac{46}{400} = 11.5\% > 10.0\%$       complaints:  $\frac{12}{400} = 3.0\% \leq 4.0\%$

#### Stuck? Three quick fixes

- **B16 does not show 3.0%.** Divide complaints by eligible appointments:  $=B14/B13$  , not by no-shows.
- **B15 shows #DIV/0!.** B13 is still empty. Type 400 in B13.
- **B15 shows 0.115.** Choose **Home > Percent Style**, then set one decimal place.

#### Why can't one month show that performance is worsening?

"Worsening" compares two periods. You have only one month, so there is nothing to compare against. Ask for an earlier month with the same KPI definition. Even then, a change over time does not show *why* it changed.

## Part B · Is MetroMeal's latest week below target?

From slide 19 → work this card → Back to slide 20

· bring: One bounded decision question, one supported descriptive claim, and one causal claim to remove.

### WHAT THE SESSION COVERED · SECTION 1 DECISION FRAME AND RUNNING-CASE BRIEF

Same frame, now with the MetroMeal numbers from the slides. The director may investigate weekend staffing before the next scheduling cycle.

- **On-time rate:** the KPI. 392 on time out of 500 eligible deliveries.
- **Target gap:** the direct difference from the 85.0% target, in **percentage points**.  $(\text{rate} - T) \times 100$
- **Complaint guardrail:** complaints among the same 500 deliveries must stay at or below 3.0%.

#### COURSE SLIDES

Slides 9–18

Decision-First Analysis 10–11 · KPI, Target, and Guardrail 12–14 · MetroMeal's Bounded Question 15–16 · Retrieval Check 18

#### CONCISE TOOLKIT

pp. 4–7

Decision-first analysis p. 4 · KPI, target and guardrail p. 7

#### FULL TOOLKIT

pp. 4–17

Decision-first analysis p. 9

Heads-up: B9 is preformatted as a percentage. Before you type the complaint count, set B9 to **General** (Home > Number). Otherwise Excel stores 14 as 14%.

### THE EQUATIONS YOU ARE BUILDING

$O$  = on-time deliveries ·  $E$  = eligible deliveries ·  $C$  = complaints ·  $T$  = target

$$\text{rate} = \frac{O}{E} = \frac{B6}{B7} \quad \text{gap}_{pp} = (B11 - B8) \times 100 \quad \text{complaint rate} = \frac{C}{E} = \frac{B9}{B7}$$

### FIRST, THE SETUP · ONCE PER SPSS SESSION

Run this **once each time you open SPSS**, before any part. On the CD line, replace

CHANGE\_TO\_YOUR\_EXTRACTED\_LAB\_FOLDER with the path of your **unzipped lab folder** (keep the single quotes and the final period). It is the folder that holds S02P\_SPSS.sps and the data files, **not** the .sps file itself: see [Find your folder path](#). Select from the first line down to the end of the H0 block and choose **Run > Selection**. Already ran it since you opened SPSS? Skip to the block for this part.

```
* =====.
* S02P practice: KPIs, variables, grain, chart critique, and bounded stories.
* MASY1-GC 1015: Quantitative Methods for Business Analysis.
* This practice is ungraded.
* =====.
```

```
CD 'CHANGE_TO_YOUR_EXTRACTED_LAB_FOLDER'. * ← change only this path
* Change an SPSS setting for this session.
SET DECIMAL=DOT.
```

*\* Import the real MetroMeal CSV used in the guided build.*

TITLE 'Import orders\_story.csv: expect 2000 eligible completed deliveries'.

*\* Read the data file into SPSS (file type, name, and each variable's format).*

GET DATA

/TYPE=TEXT

/FILE='orders\_story.csv'

/ENCODING='UTF8'

/DELCASE=LINE

/DELIMITERS=", "

/QUALIFIER='\"'

/ARRANGEMENT=DELIMITED

/FIRSTCASE=2

/VARIABLES=

order\_id F8.0

customer\_id F8.0

completed\_date SDATE10

week\_start SDATE10

weekend\_flag F1.0

region A12

item\_count F3.0

promised\_minutes F8.0

delivery\_minutes F8.2

late\_minutes F8.2

satisfaction\_category F2.0

eligible\_kpi\_flag F1.0

on\_time\_kpi\_flag F1.0

complaint\_flag F1.0

target\_rate F8.4

complaint\_guardrail F8.4.

*\* Give the open dataset a name so later commands can point to it.*

DATASET NAME OrdersStory.

*\* Set how values display (width and decimals); the stored values do not change.*

FORMATS completed\_date week\_start (SDATE10).

*\* Attach readable descriptions to variables (they appear in output tables).*

VARIABLE LABELS

order\_id 'Delivery identifier and candidate key'

customer\_id 'Customer identifier, not a measure'

completed\_date 'Date delivery was completed'

week\_start 'Start date of the reporting week'

weekend\_flag '1 for weekend completion, 0 otherwise'

region 'MetroMeal region'

item\_count 'Number of items in the order'

promised\_minutes 'Promised delivery duration in minutes'

delivery\_minutes 'Observed delivery duration in minutes'

late\_minutes 'Minutes late, with zero for an on-time delivery'

satisfaction\_category 'Ordered satisfaction category from 1 to 5'

eligible\_kpi\_flag '1 when the delivery is eligible for the KPI'

on\_time\_kpi\_flag '1 when the eligible delivery is on time'

complaint\_flag '1 when a complaint was recorded'

target\_rate 'Management on-time target'

complaint\_guardrail 'Maximum complaint-rate guardrail'.

*\* Attach readable names to codes such as 0 and 1.*

VALUE LABELS weekend\_flag 0 'Weekday' 1 'Weekend'

/eligible\_kpi\_flag 0 'Not eligible' 1 'Eligible'

/on\_time\_kpi\_flag 0 'Late' 1 'On time'

/complaint\_flag 0 'No complaint' 1 'Complaint'.

*\* Run the pending transformations now.*  
EXECUTE.

## BEFORE THE TIMER · INSTRUCTOR DEMO

Follow along: in SPSS, run the **H1** block and leave **Block H1: decision, KPI, target, and guardrail** open.

## SPSS BLOCK FOR THIS PART · ALREADY IN YOUR SYNTAX FILE

You do not type this: it is already in `S02P_SPSS.sps`, the one syntax file you use all session. Once per SPSS session, run the `CD` line (line 7) and the first block (**Import orders\_story.csv**) first (the setup code is above). Then press **Ctrl+F**, search `TITLE 'H1'`, select from the first comment line down to the last period, and choose **Run > Selection**. Green lines start with `*`: they are comments that explain the command below them, and SPSS ignores them. Every command ends with a period.

```
* Hand-off Block H1.
TITLE 'Block H1: decision, KPI, target, and guardrail'.
NEW FILE.
DATA LIST FREE / ontime eligible target complaints guardrail.
BEGIN DATA
392 500 .85 14 .03
END DATA.
* Create or overwrite a variable with a formula, row by row.
COMPUTE rate=ontime/eligible.
* Create or overwrite a variable with a formula, row by row.
COMPUTE gap_pp=(rate-target)*100.
* Create or overwrite a variable with a formula, row by row.
COMPUTE complaint_rate=complaints/eligible.
* Set how values display (width and decimals); the stored values do not change.
FORMATS rate target guardrail complaint_rate (PCT5.1) gap_pp (F6.1).
* Print the listed variables row by row.
LIST.
```

## OPEN

Open `Slide_Practice!B6`.

## FILL IN THE YELLOW CELLS

| # | Cell  | Type this     | In math                                          | Check |
|---|-------|---------------|--------------------------------------------------|-------|
| 1 | B6:B7 | 392 500       | $O, E$ counts                                    |       |
| 2 | B8    | 0.85          | $T$ target                                       | 85.0% |
| 3 | B9    | 14            | $C$ set B9 to General first                      | 14    |
| 4 | B10   | 0.03          | $G$ guardrail                                    | 3.0%  |
| 5 | B11   | =B6/B7        | $O/E$ on-time rate                               | 78.4% |
| 6 | B12   | =(B11-B8)*100 | $(\text{rate} - T) \times 100$ percentage points | -6.6  |
| 7 | B13   | =B9/B7        | $C/E$ complaint rate                             | 2.8%  |

| # | Cell | Type this                                                                                                      | In math | Check |
|---|------|----------------------------------------------------------------------------------------------------------------|---------|-------|
|   | B14  | <b>B14:</b> Should the director investigate weekend staffing before the next scheduling cycle?                 |         |       |
|   | B15  | <b>B15:</b> Name the observed target miss and the guardrail status.                                            |         |       |
| 8 | B16  | <b>B16:</b> State the staffing-cause claim the records cannot support.                                         |         |       |
|   | B17  | <b>B17:</b> One source to check before changing staffing, e.g. rosters, traffic, weather, region or order mix. |         |       |

✅ **Done when:** B11 shows **78.4%**, B12 shows **-6.6** and B13 shows **2.8%**. SPSS H1 lists the same **.784**, **-6.6** and **.028**. **Save.**

⚠️ **Most common mistake:** calling -6.6 a "6.6% drop". It is a gap of 6.6 **percentage points**. A percent change is a different calculation that gives a different number.

$$(0.784 - 0.850) \times 100 = -6.6 \text{ pp} \neq \frac{0.784 - 0.850}{0.850} \times 100 = -7.8\%$$

#### Stuck? Three quick fixes

- **B13 shows 0.0% (or B9 shows 14.0%).** B9 came preformatted as a percentage, so Excel stored 14 as 14%. Select B9, choose **Home > Number > General**, and type **14** again.
- **B12 shows -0.066.** You left out **\*100**. The gap is reported in percentage points.
- **B12 shows -6.6%.** B12 should be a plain number. Choose **Home > Number > Number** with one decimal.

## Part C · Is the KPI calculation reproducible?

From slide 38 → work this card → Back to slide 39

· bring: A checked KPI formula with units, visible numerator and denominator, target gap, and one limitation.

### WHAT THE SESSION COVERED · SECTION 2 OPERATIONAL KPI DEFINITIONS AND STATISTICAL MEANING

A KPI is **reproducible** when another analyst could compute the same number from your definition alone. Write down:

- **Population and window:** which records count, and when. Here: eligible NorthLine support calls in September 2026.
- **Numerator and denominator:** calls resolved within 5 minutes, over all eligible calls.  $\text{rate} = \frac{684}{760}$
- **Direction and target:** higher is better; the target is at least 92.0%.
- **Count or proportion:** a count shows workload; a proportion compares performance across eligible opportunities.

#### COURSE SLIDES

Slides 20–37

Reproducible KPI Specification 21–22 · Count, Proportion, and Target Gap 23–25 · Percentage Points vs Percent Change 26–27 · Reproduce the KPI in Excel 365 28 · Aggregation from Counts 32–34

#### CONCISE TOOLKIT

pp. 8–12

Reproducible KPI specification p. 8

#### FULL TOOLKIT

pp. 18–31

Reproducible KPI specification p. 31

### THE EQUATIONS YOU ARE BUILDING

$R$  = calls resolved within 5 minutes ·  $E$  = eligible calls ·  $T$  = target

$$\text{rate} = \frac{R}{E} = \frac{B21}{B22} \quad \text{gap}_{pp} = (B30 - B23) \times 100$$

#### OPEN

Open `Class_Practice!B21`.

#### FILL IN THE YELLOW CELLS

| # | Cell    | Type this        | In math       | Check      |
|---|---------|------------------|---------------|------------|
| 1 | B21:B22 | 684 760          | $R, E$ counts |            |
| 2 | B23     | 0.92             | $T$ target    | 92.0%      |
| 3 | B24     | =DATE(2026,9,1)  | Window start  | 2026-09-01 |
| 4 | B25     | =DATE(2026,9,30) | Window end    | 2026-09-30 |

| # | Cell | Type this                                                                                                                     | In math                                          | Check |
|---|------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-------|
| 5 | B30  | =B21/B22                                                                                                                      | $R/E$ resolution rate                            | 90.0% |
| 6 | B31  | =(B30-B23)*100                                                                                                                | $(\text{rate} - T) \times 100$ percentage points | -2.0  |
| 7 | B26  | <b>B26:</b> Who is in the denominator? (eligible NorthLine support calls)                                                     |                                                  |       |
|   | B27  | <b>B27:</b> Which period? (September 2026)                                                                                    |                                                  |       |
|   | B28  | <b>B28:</b> Label the qualifying count in words.                                                                              |                                                  |       |
|   | B29  | <b>B29:</b> Label the eligible count in words.                                                                                |                                                  |       |
|   | B32  | <b>B32:</b> Which one answers the service-performance question, and why?                                                      |                                                  |       |
|   | B33  | <b>B33:</b> One thing this descriptive rate cannot explain.                                                                   |                                                  |       |
|   | B34  | <b>B34:</b> One compact line: purpose, unit, population, window, grain, eligibility, calculation, direction, target, refresh. |                                                  |       |
|   | B35  | <b>B35:</b> Could a changed eligibility rule change NorthLine's decision?                                                     |                                                  |       |

✓ **Done when:** B30 shows 90.0% and B31 shows -2.0 percentage points. B34 holds a full KPI specification. Save.

⚠ **Most common mistake:** changing who is in the denominator (for example adding abandoned calls) and still calling it the same KPI. A new eligibility rule makes a new KPI version.

#### Stuck? Three quick fixes

- **B24 shows a number like 46266.** Format B24:B25 as dates: **Home > Number > Short Date**.
- **B31 shows -0.02.** Multiply the difference by 100: =(B30-B23)\*100 .
- **B30 shows 111.1%.** The division is upside down. Use =B21/B22 , qualifying over eligible.

#### Why write a full KPI specification? (optional)

The same words, "resolution rate", can hide different populations, windows and rules. The specification in B34 pins them down: purpose, unit, population, September window, one call per row, eligibility, calculation, higher-is-better, the 92.0% target and a refresh schedule.

## Part D · What is the four-week on-time rate, built from counts?

From slide 38 → work this card → Back to slide 39

· bring: A checked KPI formula with units, visible numerator and denominator, target gap, and one limitation.

### WHAT THE SESSION COVERED · SECTION 2 OPERATIONAL KPI DEFINITIONS AND STATISTICAL MEANING

To combine weeks, **add the counts first, then divide**. This is count-first aggregation.

- **Weekly rate:**  $\text{rate}_w = O_w / E_w$  for each week  $w$ .
- **Overall rate:**  $\sum O_w / \sum E_w$ . Each delivery counts once.
- **Why not average the rates?:** an average of rates gives every week equal weight, even when one week has many more deliveries.

#### COURSE SLIDES

Slides 20–37

Reproducible KPI Specification 21–22 · Count, Proportion, and Target Gap 23–25 · Percentage Points vs Percent Change 26–27 · Reproduce the KPI in Excel 365 28 · Aggregation from Counts 32–34

#### CONCISE TOOLKIT

pp. 8–12

Reproducible KPI specification p. 8

#### FULL TOOLKIT

pp. 18–31

Reproducible KPI specification p. 31

### THE EQUATIONS YOU ARE BUILDING

$O_w$  = on-time deliveries in week  $w$  ·  $E_w$  = eligible deliveries in week  $w$

$$\text{Four-week rate} = \frac{\sum_{w=1}^4 O_w}{\sum_{w=1}^4 E_w} = \frac{\text{SUM(B21:B24)}}{\text{SUM(B25:B28)}}$$

Just opened SPSS? Run the setup first (the **CD** line and H0; code in Part B).

### BEFORE THE TIMER · INSTRUCTOR DEMO

Follow along: in SPSS, run the **H2** block and leave **Block H2: weekly and aggregate KPI calculations** open.

### SPSS BLOCK FOR THIS PART · ALREADY IN YOUR SYNTAX FILE

You do not type this: it is already in **S02P\_SPSS.sps**, the one syntax file you use all session. Once per SPSS session, run the **CD** line (line 7) and the first block (**Import orders\_story.csv**) first (setup code in Part B). Then press **Ctrl+F**, search **TITLE 'H2'**, select from the first comment line down to the last period, and choose **Run > Selection**. Green lines start with **\***: they are comments that explain the command below them, and SPSS ignores them. Every command ends with a period.

*\* Hand-off Block H2.*

TITLE 'Block H2: weekly and aggregate KPI calculations'.

NEW FILE.

DATA LIST FREE / week (A10) ontime eligible target.

BEGIN DATA

2026-08-03 420 500 .85

```

2026-08-10 438 500 .85
2026-08-17 425 500 .85
2026-08-24 392 500 .85
END DATA.
* Create or overwrite a variable with a formula, row by row.
COMPUTE rate=ontime/eligible.
* Create or overwrite a variable with a formula, row by row.
COMPUTE gap_pp=(rate-target)*100.
* Set how values display (width and decimals); the stored values do not change.
FORMATS rate target (PCT5.1) gap_pp (F6.1).
* Print the listed variables row by row.
LIST.
* Collapse rows into group summaries (one row per group).
AGGREGATE
  /OUTFILE=*
  /BREAK=
  /ontime_total=SUM(ontime)
  /eligible_total=SUM(eligible).
* Create or overwrite a variable with a formula, row by row.
COMPUTE overall_rate=ontime_total/eligible_total.
* Set how values display (width and decimals); the stored values do not change.
FORMATS overall_rate (PCT6.2).
* Print the listed variables row by row.
LIST.

```

## OPEN

Open `Slide_Practice!B21`.

## FILL IN THE YELLOW CELLS

| # | Cell    | Type this                                                                                                         | In math                                     | Check  |
|---|---------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------|--------|
| 1 | B21:B24 | <code>420</code> <code>438</code> <code>425</code> <code>392</code>                                               | $O_1, \dots, O_4$ on-time counts            |        |
| 2 | B25:B28 | <code>500</code> <code>500</code> <code>500</code> <code>500</code>                                               | $E_1, \dots, E_4$ eligible counts           |        |
| 3 | B29     | <code>=B24/B28</code>                                                                                             | $O_4/E_4$ latest rate                       | 78.4%  |
| 4 | B30     | <code>0.85</code>                                                                                                 | $T$ target                                  | 85.0%  |
| 5 | B31     | <code>=(B29-B30)*100</code>                                                                                       | $(rate_4 - T) \times 100$ percentage points | -6.6   |
| 6 | B32     | <code>14</code>                                                                                                   | $C_4$ latest complaints                     | 14     |
| 7 | B33     | <code>=B32/B28</code>                                                                                             | $C_4/E_4$ complaint rate                    | 2.8%   |
| 8 | B34     | <code>=SUM(B21:B24)/SUM(B25:B28)</code>                                                                           | $\sum O_w / \sum E_w$ count-first           | 83.75% |
| 9 | B35     | <b>B35:</b> Explain why a simple average of the four rates works here only because every week has 500 deliveries. |                                             |        |

✅ **Done when:** `B29` shows 78.4%, `B33` shows 2.8% and `B34` shows 83.75% (1,675 of 2,000). SPSS H2 lists .8375 . **Save.**

⚠ **Most common mistake:** typing `=AVERAGE(...)` of the four weekly rates. It happens to give the same answer here only because every week has 500 deliveries. With unequal weeks it is wrong.

#### Stuck? Three quick fixes

- **B33 shows 0.7%.** You divided by all four weeks. The guardrail is for the latest week: `=B32/B28`.
- **B34 shows 83.8% instead of 83.75%.** Add a decimal place: **Home > Increase Decimal**.
- **B29 shows 127.6%.** The division is upside down. Use `=B24/B28`.

#### Why does count-first matter when weeks differ? (optional)

$$\frac{90\% + 75\%}{2} = 82.5\% \quad \neq \quad \frac{90 + 300}{100 + 400} = \frac{390}{500} = 78.0\%$$

Suppose week 1 has 90 of 100 on time (90.0%) and week 2 has 300 of 400 (75.0%). The average of the two rates is 82.5%. But only 390 of 500 deliveries were on time, which is 78.0%. The bigger week should carry more weight, and summing counts does that automatically.

## Part E · What does one row represent?

3 min

From slide 54 → work this card → Back to slide 55

· bring: A completed type-and-scale classification, two grain statements, and one mixed-grain warning.

### WHAT THE SESSION COVERED · SECTION 3 VARIABLES, MEASUREMENT SCALES, OBSERVATIONS, AND GRAIN

Before you count anything, finish the sentence "**Each row represents one ...**". That is the table's **grain**.

- **Data type:** categorical (labels, codes, flags) or numerical (discrete counts, continuous measures).
- **Measurement scale:** nominal (names only), ordinal (ordered), interval, or ratio (true zero, ratios make sense).
- **Analytical role:** identifier, measure, category, ordered category or flag.
- **Join risk:** joining orders (one row per order) to items (one row per item) gives one row per *item*. Orders with many items repeat.

#### COURSE SLIDES

Slides 39–53

Variables and Observations 40–41 ·  
Data Type and Measurement Scale  
42–43 · Row Grain and Supported  
Claims 45–46 · Mixed-Grain Join  
Risk 47 · Visualizing the Join Fan  
Trap 48–49 · Diagnose the repeated-  
row example 50

#### CONCISE TOOLKIT

pp. 13–16

Data type and measurement scale p.  
13

#### FULL TOOLKIT

pp. 32–44

Data type and measurement scale p.  
32

### THE EQUATIONS YOU ARE BUILDING

$$n_{\text{orders}} = \text{order rows} \cdot n_{\text{items}} = \text{item rows}$$

#### ONE-TO-MANY JOIN

$$\text{rows after joining every item to its order} = n_{\text{items}} = \text{B46}$$

### OPEN

Open Class\_Practice!C40 .

### FILL IN THE YELLOW CELLS

| # | Cell    | Type this | In math                                             | Check |
|---|---------|-----------|-----------------------------------------------------|-------|
| 1 | B45:B46 | 120 310   | $n_{\text{orders}}$ , $n_{\text{items}}$ row counts |       |
| 2 | B49     | =B46      | $n_{\text{items}}$ predicted joined rows            | 310   |

| # | Cell    | Type this                                                                                                   | In math | Check |
|---|---------|-------------------------------------------------------------------------------------------------------------|---------|-------|
|   | C40:F44 | <b>C40:F44:</b> For each field in column B, type storage, data type, measurement scale and analytical role. |         |       |
|   | B47     | <b>B47:</b> Finish: "Each row represents one ..."                                                           |         |       |
| 3 | B48     | <b>B48:</b> Finish: "Each row represents one ..."                                                           |         |       |
|   | B50     | <b>B50:</b> Which grain should the order-level return rate use, and what must you do first?                 |         |       |
|   | B51     | <b>B51:</b> What goes wrong if you calculate the order-level rate on the joined rows?                       |         |       |
|   | B52     | <b>B52:</b> Whose experience receives extra weight?                                                         |         |       |

✓ **Done when:** B49 shows 310 joined rows. B47 and B48 state one row per order and one row per item. Save.

⚠ **Most common mistake:** deleting repeated `order_id` values to "clean" the joined table. The repeats are expected at item grain. First decide which grain the question needs, then aggregate to one row per order.

#### Stuck? Three quick fixes

- **Unsure if `order_id` is numerical.** It is made of digits, but averaging IDs means nothing. It is categorical and nominal.
- **B49 shows 120.** An order-to-item join has one row per item. Use `=B46`.
- **Unsure about `satisfaction_category`.** Codes 1 to 5 have an order but no fixed spacing, so it is ordinal.

#### Whose experience gets extra weight after the join? (optional)

An order with three items appears three times, so its outcome counts three times in an order-level rate. Customers who buy more items quietly get more say. Restoring one row per order gives every order one vote again.

## Part F · How many orders hide in four joined rows?

From slide 54 → work this card → Back to slide 55

· bring: A completed type-and-scale classification, two grain statements, and one mixed-grain warning.

### WHAT THE SESSION COVERED · SECTION 3 VARIABLES, MEASUREMENT SCALES, OBSERVATIONS, AND GRAIN

MetroMeal's source file has one row per eligible completed delivery. The weekly table has one row per week. Aggregating changes what a row can support.

- **Raw grain:** 2,000 rows, one per eligible completed delivery.
- **Summary grain:** 4 rows, one per week.
- **Key test:** if the number of rows equals the number of distinct keys, the key is unique at that grain.

#### COURSE SLIDES

Slides 39–53

Variables and Observations 40–41 · Data Type and Measurement Scale 42–43 · Row Grain and Supported Claims 45–46 · Mixed-Grain Join Risk 47 · Visualizing the Join Fan Trap 48–49 · Diagnose the repeated-row example 50

#### CONCISE TOOLKIT

pp. 13–16

Data type and measurement scale p. 13

#### FULL TOOLKIT

pp. 32–44

Data type and measurement scale p. 32

### THE EQUATIONS YOU ARE BUILDING

`COUNTA` counts filled cells · `ROWS(UNIQUE(...))` counts distinct values ·  $n_{\text{rows}}$  = joined rows ·  $n_{\text{orders}}$  = distinct orders

$n_{\text{rows}} = \text{COUNTA}(\text{B47:B50})$        $n_{\text{orders}} = \text{ROWS}(\text{UNIQUE}(\text{B47:B50}))$

Just opened SPSS? Run the setup first (the `CD` line and H0; code in Part B).

### BEFORE THE TIMER · INSTRUCTOR DEMO

Follow along: in SPSS, run the **H3** block and leave **Block H3: item grain, repeated order IDs, and restoration to order grain** open.

### SPSS BLOCK FOR THIS PART · ALREADY IN YOUR SYNTAX FILE

You do not type this: it is already in `S02P_SPSS.sps`, the one syntax file you use all session. Once per SPSS session, run the `CD` line (line 7) and the first block (**Import orders\_story.csv**) first (setup code in Part B). Then press **Ctrl+F**, search `TITLE 'H3'`, select from the first comment line down to the last period, and choose **Run > Selection**. Green lines start with `*`: they are comments that explain the command below them, and SPSS ignores them. Every command ends with a period.

*\* Hand-off Block H3.*

`TITLE 'Block H3: item grain, repeated order IDs, and restoration to order grain'.`

`NEW FILE.`

`DATA LIST FREE / order_id (A1) item_id (A2) returned.`

```

BEGIN DATA
A I1 1
A I2 1
A I3 1
B I4 0
END DATA.

* Frequency table: how many cases fall in each category.
FREQUENCIES VARIABLES=order_id.

* Collapse rows into group summaries (one row per group).
AGGREGATE
  /OUTFILE=*
  /BREAK=order_id
  /returned_order=MAX(returned)
  /item_rows=N.

* Print the listed variables row by row.
LIST.

```

## OPEN

Open `Slide_Practice!B39`.

## FILL IN THE YELLOW CELLS

| # | Cell                         | Type this                                                                                                                                                                                                                                                                                                 | In math                             | Check |
|---|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-------|
| 1 | B39:B40                      | <code>2000</code> <code>4</code>                                                                                                                                                                                                                                                                          | Raw rows, weekly rows               |       |
| 2 | B47:B50                      | <code>A</code> <code>A</code> <code>A</code> <code>B</code>                                                                                                                                                                                                                                               | Order IDs                           |       |
| 3 | C47:C50                      | <code>I1</code> <code>I2</code> <code>I3</code> <code>I4</code>                                                                                                                                                                                                                                           | Item IDs                            |       |
| 4 | B51                          | <code>=COUNTA(B47:B50)</code>                                                                                                                                                                                                                                                                             | $n_{\text{rows}}$ joined rows       | 4     |
| 5 | C51                          | <code>=ROWS(UNIQUE(B47:B50))</code>                                                                                                                                                                                                                                                                       | $n_{\text{orders}}$ distinct orders | 2     |
| 6 | B41<br>B42<br>C43:F45<br>B52 | <b>B41:</b> Finish: "Each source row represents one ..."<br><b>B42:</b> Finish: "Each aggregate row represents one ..."<br><b>C43:F45:</b> Classify order_id, item_count and satisfaction_category: storage, type, scale, role.<br><b>B52:</b> What must happen before an order-level rate is calculated? |                                     |       |

✅ **Done when:** `B51` shows 4 joined rows and `C51` shows 2 distinct orders. SPSS H3 aggregates to 2 rows. Save.

⚠️ **Most common mistake:** treating the four joined rows as four orders. Order A appears three times because it has three items, so it would count three times in an order-level rate.

## Stuck? Three quick fixes

- **C51 shows #NAME?.** `UNIQUE` needs Excel 365. Check **File > Account > About Excel**.
- **C51 shows @UNIQUE.** Delete the `@` so the formula reads `=ROWS(UNIQUE(B47:B50))`.
- **C51 shows #SPILL!** Wrap `UNIQUE` inside `ROWS(...)` so the cell returns one number.

## Part G · Does the chart communicate honestly?

3 min

From slide 70 → work this card → Back to slide 71

· bring: One statistical defect, one accessibility defect, one repair, and a revised bounded title.

### WHAT THE SESSION COVERED · SECTION 4 CHART SELECTION AND HONEST STATISTICAL CRITIQUE

Check the numbers first, then the picture. An honest chart shows the checked values without exaggeration.

- **Baseline:** bars are read by length, so the bar axis starts at 0%.
- **Denominators:** show  $n$  next to each rate. The same rate can rest on very different amounts of evidence.
- **Accessibility:** never rely on red and green alone. Add a label, pattern or outline.
- **Title:** state what was observed. "Demand will continue falling" is a forecast three months cannot support.

#### COURSE SLIDES

Slides 55–69

Match the Display to the Question  
56–58 · Axes, Baselines, and  
Statistical Meaning 59–61 · Chart  
Critique and Accessibility Protocol  
62–63 · Apply the Critique in Excel  
365 64 · Bounded Chart  
Interpretation 65–66

#### CONCISE TOOLKIT

pp. 17–20

#### FULL TOOLKIT

pp. 45–58

### THE EQUATIONS YOU ARE BUILDING

$a_m$  = attendees in month  $m$  ·  $e_m$  = eligible residents in month  $m$

$$\text{rate}_m = \frac{a_m}{e_m} : \frac{\text{B56}}{\text{B57}}, \frac{\text{B58}}{\text{B59}}, \frac{\text{B60}}{\text{B61}}$$

#### OPEN

Open `Class_Practice!B56`.

#### FILL IN THE YELLOW CELLS

| # | Cell    | Type this         | In math                                 | Check |
|---|---------|-------------------|-----------------------------------------|-------|
| 1 | B56:B61 | 72 80 54 60 30 50 | $a_1, e_1, \dots, a_3, e_3$ count pairs |       |
| 2 | B62     | 0.85              | $T$ target                              | 85.0% |
| 3 | B63     | =B56/B57          | $a_1/e_1$ January                       | 90.0% |
| 4 | B64     | =B58/B59          | $a_2/e_2$ February                      | 90.0% |
| 5 | B65     | =B60/B61          | $a_3/e_3$ March                         | 60.0% |

| # | Cell | Type this                                                                                             | In math             | Check |
|---|------|-------------------------------------------------------------------------------------------------------|---------------------|-------|
| 6 | B66  | 0.50                                                                                                  | Flawed axis minimum | 50.0% |
| 7 | B67  | <b>B67:</b> Name the scale or denominator problem.                                                    |                     |       |
|   | B68  | <b>B68:</b> Name the color-only problem.                                                              |                     |       |
|   | B69  | <b>B69:</b> List the fixes: zero baseline, direct labels, denominators, contrast and a non-color cue. |                     |       |
|   | B70  | <b>B70:</b> State the observed March comparison without will.                                         |                     |       |

✓ **Done when:** B63:B65 show 90.0%, 90.0% and 60.0%. B70 states an observed comparison, not a forecast. **Save.**

⚠ **Most common mistake:** reading February's lower attendance count (54) as a lower rate. February also had fewer eligible residents, so its rate equals January's.  $\frac{72}{80} = 90.0\% = \frac{54}{60}$

#### Stuck? Three quick fixes

- **B64 shows 120.0%.** You divided by the wrong cell. February is =B58/B59 .
- **February looks worse than January.** Compare rates, not counts: 54 of 60 is the same 90.0% as 72 of 80.
- **Filling B63 down gives wrong rates.** Type each formula by hand; the pairs are two rows apart.

## Part H · What makes the MetroMeal dashboard chart misleading?

[From slide 70](#) → work this card → [Back to slide 71](#)

· bring: One statistical defect, one accessibility defect, one repair, and a revised bounded title.

### WHAT THE SESSION COVERED · SECTION 4 CHART SELECTION AND HONEST STATISTICAL CRITIQUE

The dashboard chart starts its bars at 75.0%, drops the denominators, uses red and green alone, and says performance *collapses*.

- **Truncated axis:** on a 75% to 90% axis, the 78.4% bar looks tiny next to 87.6%. On a 0% to 100% axis the difference is honest.
- **Every phrase from a checked cell:** the repaired title quotes the rate and gap you verify in B68 and B69.
- **Four points:** show an observed sequence. They do not show a stable trend or a forecast.

#### COURSE SLIDES

Slides 55–69

Match the Display to the Question  
56–58 · Axes, Baselines, and  
Statistical Meaning 59–61 · Chart  
Critique and Accessibility Protocol  
62–63 · Apply the Critique in Excel  
365 64 · Bounded Chart  
Interpretation 65–66

#### CONCISE TOOLKIT

pp. 17–20

#### FULL TOOLKIT

pp. 45–58

### THE EQUATIONS YOU ARE BUILDING

$r_4$  = latest weekly rate ·  $T$  = target

$\text{gap}_{pp} = (r_4 - T) \times 100 = (B59 - B60) \times 100$

Just opened SPSS? Run the setup first (the **CD** line and H0; code in Part B).

### BEFORE THE TIMER · INSTRUCTOR DEMO

Follow along: in SPSS, run the **H4** block and leave **Block H4: four weekly rates and a simple zero-baseline bar chart** open.

### SPSS BLOCK FOR THIS PART · ALREADY IN YOUR SYNTAX FILE

You do not type this: it is already in **S02P\_SPSS.sps**, the one syntax file you use all session. Once per SPSS session, run the **CD** line (line 7) and the first block (**Import orders\_story.csv**) first (setup code in Part B). Then press **Ctrl+F**, search **TITLE 'H4'**, select from the first comment line down to the last period, and choose **Run > Selection**. Green lines start with **\***: they are comments that explain the command below them, and SPSS ignores them. Every command ends with a period.

\* Hand-off Block H4.

TITLE 'Block H4: four weekly rates and a simple zero-baseline bar chart'.

NEW FILE.

DATA LIST FREE / week (A10) rate target eligible.

```

BEGIN DATA
2026-08-03 .840 .850 500
2026-08-10 .876 .850 500
2026-08-17 .850 .850 500
2026-08-24 .784 .850 500
END DATA.
* Set how values display (width and decimals); the stored values do not change.
FORMATS rate target (PCT5.1).
* Print the listed variables row by row.
LIST.
* Draw a chart.
GRAPH /BAR(SIMPLE)=MEAN(rate) BY week.

```

## OPEN

Open `Slide_Practice!B56`.

## FILL IN THE YELLOW CELLS

| # | Cell    | Type this                                                                                       | In math                                  | Check                      |
|---|---------|-------------------------------------------------------------------------------------------------|------------------------------------------|----------------------------|
| 1 | B56:B59 | 0.84 0.876 0.85 0.784                                                                           | $r_1, \dots, r_4$ weekly rates           | 84.0%, 87.6%, 85.0%, 78.4% |
| 2 | B60     | 0.85                                                                                            | $T$ target                               | 85.0%                      |
| 3 | B61:B62 | 0.75 0.90                                                                                       | Flawed axis                              | 75.0%, 90.0%               |
| 4 | B63     | 500                                                                                             | $E_4$ latest denominator                 |                            |
| 5 | B68     | =B59                                                                                            | $r_4$ verified latest rate               | 78.4%                      |
| 6 | B69     | =(B59-B60)*100                                                                                  | $(r_4 - T) \times 100$ percentage points | -6.6                       |
| 7 | B64     | <b>B64:</b> Name the baseline and denominator problems.                                         |                                          |                            |
|   | B65     | <b>B65:</b> Name the color-only problem.                                                        |                                          |                            |
|   | B66     | <b>B66:</b> Which word is dramatic and unsupported?                                             |                                          |                            |
|   | B67     | <b>B67:</b> 0.0% to 100.0% axis, direct labels, denominators, 85.0% target line, non-color cue. |                                          |                            |
|   | B70     | <b>B70:</b> State the latest observed rate against the 85.0% target.                            |                                          |                            |

✓ **Done when:** B68 shows 78.4% and B69 shows -6.6. SPSS H4 draws the four bars from a zero baseline. Save.

⚠ **Most common mistake:** fixing the colors but keeping the 75.0% baseline. The bars still exaggerate the dip. Cosmetic cleanup is not a statistical repair.

### Stuck? Three quick fixes

- **The 78.4% week still feels like a collapse.** On a 0% to 100% axis it sits 9.2 points below the 87.6% week. Describe that gap, not the drama.
- **B69 shows -0.066.** Multiply by 100:  $= (B59 - B60) * 100$ .
- **B68 shows 0.784.** Choose **Home > Percent Style** with one decimal.

# Part I · Is the data story traceable and bounded?

From slide 78 → work this card → Back to slide 79

· bring: One explanatory title, a three-sentence bounded story, one action, one limitation, and one next validation check.

## WHAT THE SESSION COVERED · SECTION 5 BOUNDED STORY, VALIDATION, AND BRIDGE TO DESCRIPTIVE EVIDENCE

A **bounded story** says only what the records can support. It links the audience, the checked result, the comparison, an action, a limitation and a next check.

- **Supported:** the observed rate and its gap to target, for the stated period.
- **Not supported:** "the new policy caused the miss". Three descriptive months cannot isolate a cause.
- **Bounded action:** *investigate* before *revising*.

### COURSE SLIDES

Slides 72–77  
From Exploration to a Bounded Story  
73 · When Is the Story Ready? 74 ·  
Independent Validation and Forward  
Bridge 75

### CONCISE TOOLKIT

pp. 17–20  
Bounded claim p. 19

### FULL TOOLKIT

pp. 45–58  
Bounded claim p. 54 · Independent  
validation p. 50

## THE EQUATIONS YOU ARE BUILDING

$D$  = repairs done within the window ·  $E$  = eligible repairs ·  $T$  = target

$$\text{rate} = \frac{D}{E} = \frac{B74}{B75} \quad \text{gap}_{pp} = (B77 - B76) \times 100$$

## OPEN

Open `Class_Practice!B74` . Press **Ctrl+G** (Go To), type `B74` and press **Enter**.

## FILL IN THE YELLOW CELLS

| # | Cell    | Type this      | In math                                          | Check |
|---|---------|----------------|--------------------------------------------------|-------|
| 1 | B74:B75 | 273 300        | $D, E$ counts                                    |       |
| 2 | B76     | 0.95           | $T$ target                                       | 95.0% |
| 3 | B77     | =B74/B75       | $D/E$ completion rate                            | 91.0% |
| 4 | B78     | =(B77-B76)*100 | $(\text{rate} - T) \times 100$ percentage points | -4.0  |
| 5 | B79     | 3              | Observed months                                  |       |

| # | Cell | Type this                                                                     | In math | Check |
|---|------|-------------------------------------------------------------------------------|---------|-------|
|   | B80  | <b>B80:</b> Copy the draft's causal claim.                                    |         |       |
|   | B81  | <b>B81:</b> What the checked records do show.                                 |         |       |
|   | B82  | <b>B82:</b> Name the claim the records do not establish.                      |         |       |
|   | B83  | <b>B83:</b> Investigate before revising the policy.                           |         |       |
| 6 | B84  | <b>B84:</b> Name the three-month descriptive scope.                           |         |       |
|   | B85  | <b>B85:</b> What evidence would you request next?                             |         |       |
|   | B86  | <b>B86:</b> The observed rate against the 95.0% target.                       |         |       |
|   | B87  | <b>B87:</b> Result and comparison; what it does not show; next investigation. |         |       |
|   | B88  | <b>B88:</b> What new evidence would change "investigate" to "revise"?         |         |       |

✅ **Done when:** B77 shows **91.0%** and B78 shows **-4.0**. B87 holds exactly three bounded sentences. **Save.**

⚠️ **Most common mistake:** keeping the causal sentence because the rate really did miss the target. A real miss does not tell you what caused it.

#### Stuck? Three quick fixes

- **B77 shows 0.91.** Choose **Home > Percent Style** with one decimal.
- **B78 shows -0.04.** Multiply by 100:  $= (B77 - B76) * 100$ .
- **Your story has four sentences.** Merge or cut: result and comparison; what it does not show; what to do next.

#### What evidence would support the causal claim? (optional)

You would need credible variation in who was under the policy, aligned comparison groups or periods, reliable scheduling records, and checks for other explanations. Until then, recommend an investigation.

## Part J · Can every number in the MetroMeal story be traced?

From slide 78 → work this card → Back to slide 79

· bring: One explanatory title, a three-sentence bounded story, one action, one limitation, and one next validation check.

### WHAT THE SESSION COVERED · SECTION 5 BOUNDED STORY, VALIDATION, AND BRIDGE TO DESCRIPTIVE EVIDENCE

Before a number goes into the story, check it a second way. **Validation** follows the chain from counts to rate, gap and guardrail.

- **Reconcile:** on-time plus late must equal eligible:  $392 + 108 - 500 = 0$ .
- **Recompute:** divide 14/500 yourself instead of copying 2.8% from a dashboard.
- **Keep it bounded:** four descriptive weeks can justify investigating staffing, not changing it.

#### COURSE SLIDES

Slides 72–77

From Exploration to a Bounded Story  
73 · When Is the Story Ready? 74 ·  
Independent Validation and Forward  
Bridge 75

#### CONCISE TOOLKIT

pp. 17–20

Bounded claim p. 19

#### FULL TOOLKIT

pp. 45–58

Bounded claim p. 54 · Independent  
validation p. 50

Heads-up: B77 is preformatted as a percentage. Before you type the complaint count, set B77 to **General** (Home > Number). Otherwise Excel stores 14 as 14%.

### THE EQUATIONS YOU ARE BUILDING

$O$  = on time ·  $L$  = late ·  $E$  = eligible ·  $C$  = complaints

#### RECONCILIATION

$$O + L - E = 0 \text{ rate} = \frac{B74}{B75} \quad \text{gap}_{pp} = (B79 - B76) \times 100 \quad \text{complaint rate} = \frac{B77}{B75}$$

Just opened SPSS? Run the setup first (the CD line and H0; code in Part B).

### BEFORE THE TIMER · INSTRUCTOR DEMO

Follow along: in SPSS, run the **H5** block and leave **Block H5: reconcile the latest evidence and calculate story values** open.

### SPSS BLOCK FOR THIS PART · ALREADY IN YOUR SYNTAX FILE

You do not type this: it is already in S02P\_SPSS.sps, the one syntax file you use all session. Once per SPSS session, run the CD line (line 7) and the first block (**Import orders\_story.csv**) first (setup code in Part B). Then press **Ctrl+F**, search TITLE 'H5', select from the first comment line down to the last period, and choose **Run > Selection**. Green lines start with \* : they are comments that explain the command below them, and SPSS ignores them. Every command ends with a period.

```

* Hand-off Block H5.
TITLE 'Block H5: reconcile the latest evidence and calculate story values'.
NEW FILE.
DATA LIST FREE / ontime late eligible target complaints guardrail.
BEGIN DATA
392 108 500 .85 14 .03
END DATA.
* Create or overwrite a variable with a formula, row by row.
COMPUTE reconcile=ontime+late-eligible.
* Create or overwrite a variable with a formula, row by row.
COMPUTE rate=ontime/eligible.
* Create or overwrite a variable with a formula, row by row.
COMPUTE gap_pp=(rate-target)*100.
* Create or overwrite a variable with a formula, row by row.
COMPUTE complaint_rate=complaints/eligible.
* Set how values display (width and decimals); the stored values do not change.
FORMATS rate target guardrail complaint_rate (PCT5.1) gap_pp (F6.1).
* Print the listed variables row by row.
LIST.

```

## OPEN

Open `Slide_Practice!B74` . Press **Ctrl+G** (Go To), type `B74` and press **Enter**.

## FILL IN THE YELLOW CELLS

| # | Cell    | Type this                                                                          | In math                                          | Check |
|---|---------|------------------------------------------------------------------------------------|--------------------------------------------------|-------|
| 1 | B74:B75 | 392 500                                                                            | $O, E$ counts                                    |       |
| 2 | B76     | 0.85                                                                               | $T$ target                                       | 85.0% |
| 3 | B77     | 14                                                                                 | $C$ set B77 to General first                     | 14    |
| 4 | B78     | 0.03                                                                               | $G$ guardrail                                    | 3.0%  |
| 5 | B79     | =B74/B75                                                                           | $O/E$ on-time rate                               | 78.4% |
| 6 | B80     | =(B79-B76)*100                                                                     | $(\text{rate} - T) \times 100$ percentage points | -6.6  |
| 7 | B81     | =B77/B75                                                                           | $C/E$ complaint rate                             | 2.8%  |
|   | B82     | <b>B82:</b> The latest observed rate against the target.                           |                                                  |       |
|   | B83     | <b>B83:</b> Target miss plus guardrail status.                                     |                                                  |       |
|   | B84     | <b>B84:</b> The staffing-cause claim the records cannot support.                   |                                                  |       |
| 8 | B85     | <b>B85:</b> Investigate staffing evidence before changing the schedule.            |                                                  |       |
|   | B86     | <b>B86:</b> Name the four descriptive weeks.                                       |                                                  |       |
|   | B87     | <b>B87:</b> Rosters, weekend assignments, eligibility, or a competing explanation. |                                                  |       |
|   | B88     | <b>B88:</b> Reconcile $392 + 108$ with 500, or divide 14 by 500 yourself.          |                                                  |       |

✓ **Done when:** B79 shows 78.4%, B80 shows -6.6 and B81 shows 2.8%. SPSS H5 shows `reconcile = 0` .  
Save.

⚠ **Most common mistake:** writing "the guardrail passed, so performance is fine". The complaint guardrail and the on-time target protect different outcomes. Report both.

#### Stuck? Three quick fixes

- **B81 shows 0.0% (or B77 shows 14.0%).** B77 came preformatted as a percentage, so Excel stored 14 as 14%. Set B77 to **General** and type 14 again.
- **B80 shows -0.066.** Multiply by 100:  $=(B79-B76)*100$ .
- **B79 shows 127.6%.** The division is upside down. Use  $=B74/B75$ .

## Part K · What does one row of the MetroMeal file represent?

Guided build · starts at slide 80, back to slide 82

### WHAT THE SESSION COVERED · SECTION 3 VARIABLES, MEASUREMENT SCALES, OBSERVATIONS, AND GRAIN

The guided build starts at **slide 80**. The instructor models this step. Look at `Orders_Story` first: the data sit in an Excel table named `Orders`.

- **Grain:** each row is one eligible completed delivery, August 3 to August 30, 2026.
- **Candidate key:** `order_id` is a key if rows and distinct IDs are equal:  $N_{\text{rows}} = N_{\text{distinct}}$ .
- **Eligibility:** `eligible_kpi_flag` is 1 when a delivery belongs in the denominator. Summing a 0/1 flag counts the 1s.
- **Decision frame:** write it in `Guided_Practice!B5` before you calculate anything.

#### COURSE SLIDES

Slides 39–53

Variables and Observations 40–41 · Data Type and Measurement Scale 42–43 · Row Grain and Supported Claims 45–46 · Mixed-Grain Join Risk 47 · Visualizing the Join Fan Trap 48–49 · Diagnose the repeated-row example 50 · Guided build 80

#### CONCISE TOOLKIT

pp. 13–16

Data type and measurement scale p. 13

#### FULL TOOLKIT

pp. 32–44

Data type and measurement scale p. 32

### THE EQUATIONS YOU ARE BUILDING

$N_{\text{rows}}$  = rows in `Orders` ·  $N_{\text{distinct}}$  = distinct `order_id` values ·  $f_i$  = `eligible_kpi_flag` of row  $i$  (0 or 1)

#### KEY TEST

$N_{\text{rows}} = B3 = B4 = N_{\text{distinct}} \Rightarrow$  `order_id` is unique at delivery grain

#### ELIGIBLE DELIVERIES

$$E = \sum_{i=1}^N f_i = B5$$

Just opened SPSS? Run the setup first (the `CD` line and H0; code in Part B).

### BEFORE THE TIMER · INSTRUCTOR DEMO

Follow along: in SPSS, run the **Guided Step 1** block and leave **Guided Step 1: source, candidate key, grain, and eligibility** open.

### SPSS BLOCK FOR THIS PART · ALREADY IN YOUR SYNTAX FILE

You do not type this: it is already in `S02P_SPSS.sps`, the one syntax file you use all session. Once per SPSS session, run the `CD` line (line 7) and the first block (**Import orders\_story.csv**) first (setup code in Part B). Then press **Ctrl+F**, search `TITLE 'Guided Step 1: source, c`, select from the first comment line down to the last period, and choose **Run > Selection**. Green lines start with `*`: they are comments that explain the command below them, and SPSS ignores them. Every command ends with a period.

```

* Guided Step 1.
TITLE 'Guided Step 1: source, candidate key, grain, and eligibility'.
* Switch to the named dataset; the next commands work on it.
DATASET ACTIVATE OrdersStory.
* Frequency table: how many cases fall in each category.
FREQUENCIES VARIABLES=eligible_kpi_flag.
* Make a working copy so the original data stay untouched.
DATASET COPY GrainCheck.
* Switch to the named dataset; the next commands work on it.
DATASET ACTIVATE GrainCheck.
* Collapse rows into group summaries (one row per group).
AGGREGATE
  /OUTFILE=*
  /BREAK=order_id
  /rows_per_order=N.
* Summary statistics (N, mean, standard deviation, minimum, maximum).
DESCRIPTIVES VARIABLES=rows_per_order /STATISTICS=SUM MIN MAX.
* Switch to the named dataset; the next commands work on it.
DATASET ACTIVATE OrdersStory.

```

## OPEN

Open `KPI_Build!B3`.

## FILL IN THE YELLOW CELLS

| # | Cell               | Type this                                                                                                                                                                                                                                         | In math                                  | Check |
|---|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------|
| 1 | B3                 | <code>=ROWS(Orders[order_id])</code>                                                                                                                                                                                                              | $N_{\text{rows}}$ all delivery rows      | 2000  |
| 2 | B4                 | <code>=ROWS(UNIQUE(Orders[order_id]))</code>                                                                                                                                                                                                      | $N_{\text{distinct}}$ distinct order IDs | 2000  |
| 3 | B5                 | <code>=SUM(Orders[eligible_kpi_flag])</code>                                                                                                                                                                                                      | $\sum f_i$ eligible deliveries           | 2000  |
| 4 | Guided_Practice!B5 | <b>Guided_Practice!B5:</b> One sentence: the regional operations director, investigating weekend staffing before the next scheduling cycle, the weekly on-time KPI, the 85.0% target, the 3.0% complaint guardrail, and no causal staffing claim. |                                          |       |

✅ **Done when:** `B3`, `B4` and `B5` each show **2000**, so `order_id` is a candidate key and every row is eligible. Save.

⚠️ **Most common mistake:** sorting or deleting rows in `Orders_Story`. The guided formulas use the whole `Orders` table. Never edit the source rows.

## Stuck? Three quick fixes

- **#NAME? in B3.** The table must be named `Orders`. Click inside `Orders_Story`, open **Table Design**, and check the name.
- **B4 shows =@UNIQUE.** Delete the `@` so the formula reads `=ROWS(UNIQUE(Orders[order_id]))`.
- **B4 shows #SPILL!** Keep `UNIQUE` inside `ROWS(...)` so the cell returns one number.

## Part L · How did each week compare with the target and the guardrail?

🕒 4 min

Guided build · starts at slide 80, back to slide 82

### WHAT THE SESSION COVERED · SECTION 2 OPERATIONAL KPI DEFINITIONS AND STATISTICAL MEANING

Build one row per week (rows 7 to 10), a count-first total (row 11) and late-delivery checks for the latest week (row 18).

- **Weekly rate:**  $\text{rate}_w = O_w / E_w$ , with the eligible count in column C and the on-time count in column D.
- **Target gap:**  $(\text{rate}_w - T) \times 100$  in percentage points.  $\$B\$14$  stays fixed when you fill down.
- **Guardrail:**  $C_w / E_w$ , complaints over the same eligible deliveries.
- **Late minutes:** the mean uses every delay; the median is the middle; the 90th percentile shows the long tail.

A mean above the median is a reason to look at the distribution. It is not proof of a cause.

#### COURSE SLIDES

Slides 20–37

Reproducible KPI Specification 21–22 · Count, Proportion, and Target Gap 23–25 · Percentage Points vs Percent Change 26–27 · Reproduce the KPI in Excel 365 28 · Aggregation from Counts 32–34 · Guided build 80

#### CONCISE TOOLKIT

pp. 8–12

Reproducible KPI specification p. 8

#### FULL TOOLKIT

pp. 18–31

Reproducible KPI specification p. 31

### THE EQUATIONS YOU ARE BUILDING

$w$  = week ·  $E_w$  = eligible ·  $O_w$  = on time ·  $C_w$  = complaints ·  $T$  = target (B14) ·  $G$  = guardrail (B15)

$$\text{rate}_w = \frac{O_w}{E_w} = \frac{D_w}{C_w} \quad \text{gap}_w = (\text{rate}_w - T) \times 100 \quad \text{complaint rate}_w = \frac{C_w}{E_w}$$

#### FOUR-WEEK AGGREGATE (ROW 11)

$$\text{rate} = \frac{\sum_w O_w}{\sum_w E_w} = \frac{D11}{C11}$$

#### RECONCILIATION (G18)

$$(E_4 - O_4) - n_{\text{late}} = 0$$

Just opened SPSS? Run the setup first (the  $\text{CD}$  line and H0; code in Part B).

### BEFORE THE TIMER · INSTRUCTOR DEMO

Follow along: in SPSS, run the **Guided Step 2** block and leave **Guided Step 2: weekly KPI, guardrail, aggregate, and late-delivery checks** open.

## SPSS BLOCK FOR THIS PART · ALREADY IN YOUR SYNTAX FILE

You do not type this: it is already in `S02P_SPSS.sps`, the one syntax file you use all session. Once per SPSS session, run the `CD` line (line 7) and the first block (**Import orders\_story.csv**) first (setup code in Part B). Then press **Ctrl+F**, search `TITLE 'Guided Step 2: weekly KP`, select from the first comment line down to the last period, and choose **Run >** **Selection**. Green lines start with `*`: they are comments that explain the command below them, and SPSS ignores them. Every command ends with a period.

```
* Guided Step 2.
TITLE 'Guided Step 2: weekly KPI, guardrail, aggregate, and late-delivery checks'.
* Switch to the named dataset; the next commands work on it.
DATASET ACTIVATE OrdersStory.
* Create an empty named dataset to receive results.
DATASET DECLARE WeeklyKPI.
* Collapse rows into group summaries (one row per group).
AGGREGATE
  /OUTFILE='WeeklyKPI'
  /BREAK=week_start
  /eligible=SUM(eligible_kpi_flag)
  /on_time=SUM(on_time_kpi_flag)
  /complaints=SUM(complaint_flag)
  /target=MAX(target_rate)
  /guardrail=MAX(complaint_guardrail).
* Switch to the named dataset; the next commands work on it.
DATASET ACTIVATE WeeklyKPI.
* Create or overwrite a variable with a formula, row by row.
COMPUTE rate=on_time/eligible.
* Create or overwrite a variable with a formula, row by row.
COMPUTE gap_pp=(rate-target)*100.
* Create or overwrite a variable with a formula, row by row.
COMPUTE complaint_rate=complaints/eligible.
* Set how values display (width and decimals); the stored values do not change.
FORMATS rate target guardrail complaint_rate (PCT5.1) gap_pp (F6.1).
* Sort the rows by the listed variables.
SORT CASES BY week_start.
* Print the listed variables row by row.
LIST VARIABLES=week_start eligible on_time rate gap_pp complaints complaint_rate target guardrail.
* Collapse rows into group summaries (one row per group).
AGGREGATE
  /OUTFILE=*
  /BREAK=
  /eligible_total=SUM(eligible)
  /on_time_total=SUM(on_time)
  /complaints_total=SUM(complaints)
  /target=MAX(target)
  /guardrail=MAX(guardrail).
* Create or overwrite a variable with a formula, row by row.
COMPUTE overall_rate=on_time_total/eligible_total.
* Create or overwrite a variable with a formula, row by row.
COMPUTE overall_gap_pp=(overall_rate-target)*100.
* Create or overwrite a variable with a formula, row by row.
COMPUTE overall_complaint_rate=complaints_total/eligible_total.
* Set how values display (width and decimals); the stored values do not change.
FORMATS overall_rate overall_complaint_rate target guardrail (PCT6.2) overall_gap_pp (F6.2).
* Print the listed variables row by row.
LIST.
* Switch to the named dataset; the next commands work on it.
```

```

DATASET ACTIVATE OrdersStory.
* Make the next transformation apply to the next procedure only.
TEMPORARY.
* Keep only the rows that meet the condition (others are deleted).
SELECT IF (week_start=DATE.DMY(24,8,2026) AND eligible_kpi_flag=1 AND late_minutes>0).
* Explore a variable: percentiles, spread and plots, optionally by group.
EXAMINE VARIABLES=late_minutes
/STATISTICS=DESCRIPTIVES
/PERCENTILES(50,90)
/MISSING=LISTWISE.

* Rebuild WeeklyKPI because the aggregate listing above reduced its active copy.
DATASET ACTIVATE OrdersStory.
* Close a dataset you no longer need.
DATASET CLOSE WeeklyKPI.
* Create an empty named dataset to receive results.
DATASET DECLARE WeeklyKPI.
* Collapse rows into group summaries (one row per group).
AGGREGATE
/OUTFILE='WeeklyKPI'
/BREAK=week_start
/eligible=SUM(eligible_kpi_flag)
/on_time=SUM(on_time_kpi_flag)
/complaints=SUM(complaint_flag)
/target=MAX(target_rate)
/guardrail=MAX(complaint_guardrail).
* Switch to the named dataset; the next commands work on it.
DATASET ACTIVATE WeeklyKPI.
* Create or overwrite a variable with a formula, row by row.
COMPUTE rate=on_time/eligible.
* Create or overwrite a variable with a formula, row by row.
COMPUTE gap_pp=(rate-target)*100.
* Create or overwrite a variable with a formula, row by row.
COMPUTE complaint_rate=complaints/eligible.
* Set how values display (width and decimals); the stored values do not change.
FORMATS rate target guardrail complaint_rate (PCT5.1) gap_pp (F6.1).
* Sort the rows by the listed variables.
SORT CASES BY week_start.
* Run the pending transformations now.
EXECUTE.

```

## OPEN

Open `KPI_Build!B14`.

## FILL IN THE YELLOW CELLS

| # | Cell | Type this                                      | In math            | Check      |
|---|------|------------------------------------------------|--------------------|------------|
| 1 | B14  | <code>=MAX(Orders[target_rate])</code>         | $T$ target         | 85.0%      |
| 2 | B15  | <code>=MAX(Orders[complaint_guardrail])</code> | $G$ guardrail      | 3.0%       |
| 3 | B7   | <code>=DATE(2026,8,3)</code>                   | $w = 1$ first week | 2026-08-03 |

| #  | Cell | Type this                                                                                                        | In math                                  | Check                              |
|----|------|------------------------------------------------------------------------------------------------------------------|------------------------------------------|------------------------------------|
| 4  | B8   | <code>=B7+7</code><br>Fill to B10.                                                                               | $w = 2, 3, 4$ next weeks                 | 2026-08-10, 2026-08-17, 2026-08-24 |
| 5  | C7   | <code>=COUNTIFS(Orders[week_start],B7,Orders[eligible_kpi_flag],1)</code><br>Fill to C10.                        | $E_w$ eligible                           | 500, 500, 500, 500                 |
| 6  | D7   | <code>=SUMIFS(Orders[on_time_kpi_flag],Orders[week_start],B7,Orders[eligible_kpi_flag],1)</code><br>Fill to D10. | $O_w$ on time                            | 420, 438, 425, 392                 |
| 7  | E7   | <code>=D7/C7</code><br>Fill to E10.                                                                              | $O_w/E_w$ on-time rate                   | 84.0%, 87.6%, 85.0%, 78.4%         |
| 8  | F7   | <code>=(E7-\$B\$14)*100</code><br>Fill to F10.                                                                   | $(rate_w - T) \times 100$<br>gap, points | -1.0, 2.6, 0.0, -6.6               |
| 9  | G7   | <code>=SUMIFS(Orders[complaint_flag],Orders[week_start],B7,Orders[eligible_kpi_flag],1)</code><br>Fill to G10.   | $C_w$ complaints                         | 10, 11, 12, 14                     |
| 10 | H7   | <code>=G7/C7</code><br>Fill to H10.                                                                              | $C_w/E_w$ complaint rate                 | 2.0%, 2.2%, 2.4%, 2.8%             |
| 11 | B11  | Four-week aggregate                                                                                              | Row label                                |                                    |
| 12 | C11  | <code>=SUM(C7:C10)</code><br>Fill to D11.                                                                        | $\sum E_w, \sum O_w$<br>summed counts    | 2000, 1675                         |
| 13 | E11  | <code>=D11/C11</code>                                                                                            | $\sum O_w / \sum E_w$ count-first rate   | 83.75%                             |
| 14 | F11  | <code>=(E11-\$B\$14)*100</code>                                                                                  | $(rate - T) \times 100$<br>exactly -1.25 | -1.3                               |
| 15 | G11  | <code>=SUM(G7:G10)</code>                                                                                        | $\sum C_w$ complaints                    | 47                                 |
| 16 | H11  | <code>=G11/C11</code>                                                                                            | $\sum C_w / \sum E_w$<br>complaint rate  | 2.35%                              |
| 17 | B18  | Latest late-delivery checks                                                                                      | Row label                                |                                    |
| 18 | C18  | <code>=C10-D10</code>                                                                                            | $E_4 - O_4$ late deliveries              | 108                                |

| #  | Cell | Type this                                                                                                                                          | In math                                      | Check                               |
|----|------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-------------------------------------|
| 19 | D18  | <code>=AVERAGEIFS(Orders[late_minutes],Orders[week_start],B10,Orders[late_minutes],"&gt;0",Orders[eligible_kpi_flag],1)</code>                     | $\bar{x}$ mean minutes late                  | 12.4                                |
| 20 | E18  | <code>=MEDIAN(FILTER(Orders[late_minutes],(Orders[week_start]=B10)*(Orders[late_minutes]&gt;0)*(Orders[eligible_kpi_flag]=1))))</code>             | $\tilde{x}$ median minutes late              | 7                                   |
| 21 | F18  | <code>=PERCENTILE.INC(FILTER(Orders[late_minutes],(Orders[week_start]=B10)*(Orders[late_minutes]&gt;0)*(Orders[eligible_kpi_flag]=1))),0.9)</code> | $x_{(0.90)}$ 90th percentile                 | 31                                  |
| 22 | G18  | <code>=C10-D10-C18</code>                                                                                                                          | $(E_4 - O_4) - n_{\text{late}}$<br>must be 0 | 0                                   |
| 23 | H18  | <code>=IF(AND(E10&lt;=\$B\$14,H10&lt;=\$B\$15),"Investigate, guardrail within limit","Review KPI and guardrail")</code>                            | Decision check                               | Investigate, guardrail within limit |

✓ **Done when:** Row 10 shows 500, 392, 78.4%, -6.6, 14 and 2.8%; E11 shows 83.75%; G18 shows 0. SPSS Guided Step 2 lists the same latest row. **Save.**

⚠ **Most common mistake:** using 392 as the denominator of the complaint rate. The guardrail is complaints among **all 500** eligible deliveries, not only the on-time ones.  $\frac{14}{500} = 2.8\% \neq \frac{14}{392} = 3.6\%$

#### Stuck? Three quick fixes

- **Every weekly count is 0.** B7:B10 must be real dates. Use `=DATE(2026,8,3)` in B7 and `=B7+7` below it.
- **F8:F10 are wrong after filling.** Lock the target: `$B$14`, not `B14`.
- **E18 or F18 shows #CALC!.** No rows passed the filter. Check that B10 shows 2026-08-24 and the condition is `late_minutes>0`.

### Why is F11 about -1.25 when the latest week is -6.6? (optional)

$$\frac{420 + 438 + 425 + 392}{2000} = \frac{1675}{2000} = 83.75\%$$

F11 pools all four weeks: 1,675 of 2,000 deliveries were on time, 83.75%. Three strong weeks hide the weak latest week. That is why the director needs the weekly rows, not only the total.

## Part M · Does the chart show the checked numbers honestly?

Guided build · starts at slide 80, back to slide 82

### WHAT THE SESSION COVERED · SECTION 4 CHART SELECTION AND HONEST STATISTICAL CRITIQUE

Link the chart table to `KPI_Build` so the chart updates when a number is fixed. Then build a column-and-line combo chart.

- **Linked, not typed:** every source cell starts with `=KPI_Build!`.
- **Zero baseline:** set the vertical axis to Minimum `0` and Maximum `1` (0% to 100%).
- **Direct labels:** label each bar with its rate and  $n$ , using the text built in `E5:E8`.
- **Non-color cue:** give the latest bar a pattern fill or a thick outline, so it shows in grayscale.

To build the chart: select `A4:C8`, choose **Insert > Combo Chart > Clustered Column: Line**. Make On-time rate the columns and Target the line, both on the primary axis. For labels, choose **Format Data Labels > Value From Cells**, select `E5:E8`, and clear the plain Value box.

#### COURSE SLIDES

Slides 55–69

Match the Display to the Question  
56–58 · Axes, Baselines, and  
Statistical Meaning 59–61 · Chart  
Critique and Accessibility Protocol  
62–63 · Apply the Critique in Excel  
365 64 · Bounded Chart  
Interpretation 65–66 · Guided build  
80

#### CONCISE TOOLKIT

pp. 17–20

#### FULL TOOLKIT

pp. 45–58

### THE EQUATIONS YOU ARE BUILDING

$r_w$  = weekly on-time rate ·  $T$  = target ·  $E_w$  = weekly denominator

#### DIRECT LABEL (E5)

$\text{label}_w = \text{TEXT}(r_w, "0.0\%") \& " (n = "&E_w \& ")"$

Just opened SPSS? Run the setup first (the `CD` line and H0; code in Part B).

### BEFORE THE TIMER · INSTRUCTOR DEMO

Follow along: in SPSS, run the **Guided Step 3** block and leave **Guided Step 3: weekly on-time chart from checked values** open.

### SPSS BLOCK FOR THIS PART · ALREADY IN YOUR SYNTAX FILE

You do not type this: it is already in `S02P_SPSS.sps`, the one syntax file you use all session. Once per SPSS session, run the `CD` line (line 7) and the first block (**Import orders\_story.csv**) first (setup code in Part B). Then press **Ctrl+F**, search `TITLE 'Guided Step 3: weekly on`, select from the first comment line down to the last period, and choose **Run > Selection**. Green lines start with `*`: they are comments that explain the command below them, and SPSS ignores them. Every command ends with a period.

```
* Guided Step 3.
TITLE 'Guided Step 3: weekly on-time chart from checked values'.
* Switch to the named dataset; the next commands work on it.
DATASET ACTIVATE WeeklyKPI.
* Draw a chart.
GRAPH /BAR(SIMPLE)=MEAN(rate) BY week_start.
```

OPEN

Open `Chart_Build!A5`.

FILL IN THE YELLOW CELLS

| # | Cell                              | Type this                                                                                                                                                                                                                                                                                                                                                                                                                     | In math            | Check                               |
|---|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------------|
| 1 | A5                                | <code>=KPI_Build!B7</code><br>Fill to A8.                                                                                                                                                                                                                                                                                                                                                                                     | $w$ week start     | 2026-08-03 ... 2026-08-24           |
| 2 | B5                                | <code>=KPI_Build!E7</code><br>Fill to B8.                                                                                                                                                                                                                                                                                                                                                                                     | $r_w$ on-time rate | 84.0%, 87.6%, 85.0%, 78.4%          |
| 3 | C5                                | <code>=KPI_Build!\$B\$14</code><br>Fill to C8.                                                                                                                                                                                                                                                                                                                                                                                | $T$ target line    | 85.0% each                          |
| 4 | D5                                | <code>=KPI_Build!C7</code><br>Fill to D8.                                                                                                                                                                                                                                                                                                                                                                                     | $E_w$ denominator  | 500 each                            |
| 5 | E5                                | <code>=TEXT(B5, "0.0%") &amp; " (n = "&amp;D5&amp;")"</code><br>Fill to E8.                                                                                                                                                                                                                                                                                                                                                   | Direct label       | 84.0% (n = 500) ... 78.4% (n = 500) |
| 6 | A4:C8<br>B12<br>B13<br>B14<br>B15 | <b>A4:C8:</b> Insert the combo chart, set the axis to 0 and 1, add labels from E5:E8, and mark the latest bar with a pattern or outline.<br><b>B12:</b> What was wrong with the flawed chart's scale and denominators?<br><b>B13:</b> Why do red and green alone fail?<br><b>B14:</b> List the repairs you made.<br><b>B15:</b> Latest observed rate against the 85.0% target, plus the limitation of four descriptive weeks. |                    |                                     |

✓ **Done when:** `B8` shows **78.4%**, `E8` shows **78.4% (n = 500)**, and the chart axis runs from 0% to 100% with an 85.0% target line. **Save.**

⚠ **Most common mistake:** typing the rates into the chart table instead of linking them. If a `KPI_Build` formula is repaired later, a typed table keeps the old numbers and the chart disagrees with the analysis.

### Stuck? Three quick fixes

- **The Combo Chart option is missing.** Insert a clustered column chart, right-click the Target series, choose **Change Series Chart Type > Line**.
- **Labels show only percentages.** Use **Value From Cells** with `E5:E8` , then clear the Value box.
- **C6:C8 point to the wrong cell.** Lock the target: `=KPI_Build!$B$14` .

## Part N · What should the director do next?

Guided build · starts at slide 80, back to slide 82

### WHAT THE SESSION COVERED · SECTION 5 BOUNDED STORY, VALIDATION, AND BRIDGE TO DESCRIPTIVE EVIDENCE

Finish by checking, then writing. Read `Self_Check` by its labels, repair any cell it flags, then write exactly **three sentences** in `Response!B7`.

- **Sentence 1:** what the evidence shows: 392 of 500, 78.4%, -6.6 points against 85.0%, and 14 of 500 complaints (2.8%) within the 3.0% guardrail.
- **Sentence 2:** what it does not show: four descriptive weeks do not show that weekend staffing caused the result or predict the next cycle.
- **Sentence 3:** what to do next: investigate delivery and staffing evidence before changing the schedule, and name one check.

Optional: enter your Excel and SPSS values side by side in `Compare_Tools` to confirm the two tools agree. After this card, return to the slides at **slide 82** for the group studio.

#### COURSE SLIDES

Slides 72–77

From Exploration to a Bounded Story  
73 · When Is the Story Ready? 74 ·  
Independent Validation and Forward  
Bridge 75 · Guided build 80

#### CONCISE TOOLKIT

pp. 17–20

Bounded claim p. 19

#### FULL TOOLKIT

pp. 45–58

Bounded claim p. 54 · Independent  
validation p. 50

### THE EQUATIONS YOU ARE BUILDING

$O$  = on time ·  $L$  = late ·  $E$  = eligible

#### THE EVIDENCE CHAIN

2000 rows  $\rightarrow \frac{392}{500} = 78.4\% \rightarrow -6.6 \text{ pp} \rightarrow \frac{14}{500} = 2.8\% \rightarrow \text{chart} \rightarrow 3 \text{ sentences}$

Just opened SPSS? Run the setup first (the `CD` line and `H0`; code in Part B).

### BEFORE THE TIMER · INSTRUCTOR DEMO

Follow along: in SPSS, run the **Guided Step 4** block and leave **Guided Step 4: values to validate against the chart title and bounded story** open.

### SPSS BLOCK FOR THIS PART · ALREADY IN YOUR SYNTAX FILE

You do not type this: it is already in `S02P_SPSS.sps`, the one syntax file you use all session. Once per SPSS session, run the `CD` line (line 7) and the first block (`Import orders_story.csv`) first (setup code in Part B). Then press **Ctrl+F**, search `TITLE 'Guided Step 4: values to`, select from the first comment line down to the last period, and choose **Run > Selection**. Green lines start with `*`: they are comments that explain the command below them, and SPSS ignores them. Every command ends with a period.

```
* Guided Step 4.
TITLE 'Guided Step 4: values to validate against the chart title and bounded story'.
* Switch to the named dataset; the next commands work on it.
DATASET ACTIVATE WeeklyKPI.
* Print the listed variables row by row.
LIST VARIABLES=week_start eligible on_time rate gap_pp complaints complaint_rate target guardrail.
```

OPEN

Open `Response!B7` .

FILL IN THE YELLOW CELLS

| # | Cell                    | Type this                                                                                                                                                                             | In math | Check |
|---|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|
| 1 | Self_Check!B5:B29<br>B7 | <b>Self_Check!B5:B29:</b> Read each result. Repair any × in the analysis cell named in column C.<br><b>B7:</b> Evidence shown; what it does not show; the next step for the director. |         |       |

✔ **Done when:** Every completed `Self_Check` row shows ✔ and `Response!B7` holds exactly three bounded sentences. **Save.**

⚠ **Most common mistake:** treating a row of ✔ marks as proof that the reasoning is sound. `Self_Check` compares numbers and counts words. It cannot tell whether a sentence claims a cause.

Stuck? Three quick fixes

- **A Self\_Check row shows ×.** Go to the cell it names in column C and fix the formula there. Never type over `Self_Check`.
- **The word-count check shows ×.** B9 counts words in B7. Aim for 30 to 200 words in three sentences.
- **Your sentence says staffing *caused* the dip.** Rewrite it as something to investigate, not a finding.

## Finish

Ungraded practice

### Before you close the workbook

TICK EVERY BOX 0 of 80 done

#### Save your work

- ☐ Workbook saved as  
S02P\_FirstName\_LastName\_NetID.xlsx
- ☐ Nothing to submit: this lab is ungraded practice

#### Part B · Slide\_Practice

- ☐ B11 shows **78.4%**, B12 shows **-6.6** and B13 shows **2.8%**. SPSS H1 lists the same **.784**, **-6.6** and **.028**.
- ☐ B14 written: Bounded decision question
- ☐ B15 written: Supported claim
- ☐ B16 written: Unsupported claim
- ☐ B17 written: Additional evidence

#### Part A · Class\_Practice

- ☐ B15 shows **11.5%** and B16 shows **3.0%**. B17 has a supported claim, a claim to remove and a revised question.
- ☐ B6 written: Stakeholder
- ☐ B7 written: Plausible decision
- ☐ B8 written: Action horizon
- ☐ B9 written: KPI
- ☐ B17 written: Claims and revised question

#### Part C · Class\_Practice

- ☐ B30 shows **90.0%** and B31 shows **-2.0** percentage points. B34 holds a full KPI specification.
- ☐ B26 written: Population
- ☐ B27 written: Reporting window
- ☐ B28 written: Numerator
- ☐ B29 written: Denominator
- ☐ B32 written: Count or proportion?
- ☐ B33 written: Limitation
- ☐ B34 written: KPI specification
- ☐ B35 written: Reflective question

#### Part D · Slide\_Practice

- ☐ B29 shows **78.4%**, B33 shows **2.8%** and B34 shows **83.75%** (1,675 of 2,000). SPSS H2 lists **.8375**.
- ☐ B35 written: Weighting limitation

#### Part E · Class\_Practice

- ☐ B49 shows **310** joined rows. B47 and B48 state one row per order and one row per item.
- ☐ C40:F44 written: Field classification
- ☐ B47 written: Orders-table grain
- ☐ B48 written: Items-table grain
- ☐ B50 written: Safe grain for the KPI
- ☐ B51 written: Mixed-grain warning
- ☐ B52 written: Reflective answer

#### Part F · Slide\_Practice

- ☐ B51 shows **4** joined rows and C51 shows **2** distinct orders. SPSS H3 aggregates to 2 rows.
- ☐ B41 written: Raw-data grain
- ☐ B42 written: Summary grain
- ☐ C43:F45 written: Field classification
- ☐ B52 written: Mixed-grain warning

#### Part G · Class\_Practice

- ☐ B63:B65 show **90.0%**, **90.0%** and **60.0%**. B70 states an observed comparison, not a forecast.
- ☐ B67 written: Statistical defect
- ☐ B68 written: Accessibility defect
- ☐ B69 written: Repair
- ☐ B70 written: Bounded title

#### Part H · Slide\_Practice

- ☐ B68 shows **78.4%** and B69 shows **-6.6**. SPSS H4 draws the four bars from a zero baseline.
- ☐ B64 written: Statistical defect
- ☐ B65 written: Accessibility defect
- ☐ B66 written: Title defect
- ☐ B67 written: Repair
- ☐ B70 written: Revised bounded title

#### Part I · Class\_Practice

- ☐ B77 shows **91.0%** and B78 shows **-4.0**. B87 holds exactly three bounded sentences.
- ☐ B80 written: Original claim
- ☐ B81 written: Supported evidence
- ☐ B82 written: Unsupported claim
- ☐ B83 written: Bounded action
- ☐ B84 written: Limitation
- ☐ B85 written: Next validation check
- ☐ B86 written: Bounded title
- ☐ B87 written: Three-sentence story
- ☐ B88 written: Reflective answer

### Part J · Slide\_Practice

- ☐ B79 shows **78.4%**, B80 shows **-6.6** and B81 shows **2.8%**. SPSS H5 shows `reconcile = 0`.
- ☐ B82 written: Bounded title
- ☐ B83 written: Evidence sentence
- ☐ B84 written: Unsupported-cause sentence
- ☐ B85 written: Recommended action
- ☐ B86 written: Evidence limitation
- ☐ B87 written: Next validation check
- ☐ B88 written: How one number was verified

### Part L · KPI\_Build

- ☐ Row 10 shows **500**, **392**, **78.4%**, **-6.6**, **14** and **2.8%**; E11 shows **83.75%**; G18 shows **0**. SPSS Guided Step 2 lists the same latest row.

### Part N · Response

- ☐ Every completed `Self_Check` row shows ✓ and `Response!B7` holds exactly three bounded sentences.
- ☐ `Self_Check!B5:B29` written: Self-check results
- ☐ B7 written: Your three sentences

### Part K · KPI\_Build

- ☐ B3, B4 and B5 each show **2000**, so `order_id` is a candidate key and every row is eligible.
- ☐ `Guided_Practice!B5` written: Decision frame

### Part M · Chart\_Build

- ☐ B8 shows **78.4%**, E8 shows **78.4% (n = 500)**, and the chart axis runs from 0% to 100% with an 85.0% target line.
- ☐ A4:C8 written: Chart
- ☐ B12 written: Statistical defect
- ☐ B13 written: Accessibility defect
- ☐ B14 written: Repair to make
- ☐ B15 written: Bounded chart title

### Whole workbook

- ☐ No `#NAME?`, `#VALUE!`, `#REF!`, `#NUM!` or `#DIV/0!` anywhere
- ☐ Data sheets unchanged; formulas, not pasted values
- ☐ Read the `Self_Check` sheet and fix anything it flags

## Optional practice (not graded)

Everything here is optional and ungraded. Each topic asks one question and answers it in Excel, then SPSS, then Python, then R. Pick the tools you want. **Topic 1** checks the numbers you already have. Topics 2 to 5 go one step beyond the lab. Put any code file in the same folder as `orders_story.csv`. Run the import block of `S02P_SPSS.sps` first, so `OrdersStory` is open. Start when you finish the practice.

Companion notebooks in your package:

- `S02P_Python_Exploratory.ipynb`
- `S02P_R_Exploratory.ipynb`

### Topic 1 · Reproduce the practice [Reproduce](#)

Same question in Excel → SPSS → Python → R.

**Excel** Parts A, B, C, D, E, F, G, H, I, J, K, L, M, N

No code needed: your workbook is the Excel version. Open **Self\_Check**. Every row should show a tick. Then check these numbers in `KPI_Build`: 2,000 rows, latest on-time rate 78.4%, gap -6.6 points, complaint rate 2.8%, four-week rate 83.75%, and in row 18: 108 late deliveries with mean 12.4, median 7 and 90th percentile 31 minutes. Type your Excel values into column C of **Compare\_Tools**, rows 7 to 13.

**SPSS** Parts A, B, C, D, E, F, G, H, I, J, K, L, M, N

No new code. Run `S02P_SPSS.sps` one block at a time. **Guided Step 1** should show 2,000 eligible rows. **Guided Step 2** lists the weekly table (latest 78.4%, gap -6.6) and the four-week aggregate (83.75%), then describes the 108 late deliveries. **Hand-off Blocks H1 to H5** redo the typed practice cases. Type the SPSS values into column D of **Compare\_Tools**. SPSS uses a different percentile rule from `PERCENTILE.INC`, so its 90th percentile can differ a little from 31.

Recomputes every practice rate, the weekly KPI table, the count-first aggregate, the late-delivery checks and the chart labels. It also saves the repaired Part M chart as `S02P_repaired_chart.png` : zero baseline, labels with n, a dashed target line and a hatch pattern on the latest bar. Your printed numbers should match the workbook: 78.4%, -6.6, 2.8%, 83.75%, 108 late, 12.4 / 7.0 / 31.0 minutes. Runs in Colab or any Python with pandas and matplotlib.

```
# S02P Parts A-N in Python (optional, not graded).
# Put this file next to orders_story.csv.
import matplotlib
matplotlib.use("Agg")          # save the chart to a file; no window needed
import matplotlib.pyplot as plt
from matplotlib.ticker import PercentFormatter
import pandas as pd

def pct(x, d=1):
    return f"{x:.{d}%}"

def gap(rate, target):          # percentage points, not percent change
    return (rate - target) * 100

# ---- Parts A-J: the five practice cases (typed numbers) ----
print("A HarborCare no-show rate =", pct(46 / 400), "| complaint rate =", pct(12 / 400))
print("B MetroMeal on-time rate =", pct(392 / 500),
      "| gap =", f"{gap(392 / 500, 0.85):.1f}", "pp | complaint rate =", pct(14 / 500))
print("C NorthLine resolution =", pct(684 / 760),
      "| gap =", f"{gap(684 / 760, 0.92):.1f}", "pp")
weeks_on_time, weeks_eligible = [420, 438, 425, 392], [500, 500, 500, 500]
print("D MetroMeal four-week rate =", pct(sum(weeks_on_time) / sum(weeks_eligible), 2),
      "(sum counts, then divide)")
print("E BrightCart joined rows =", 310, "item rows for", 120, "orders")
items = pd.DataFrame({"order_id": ["A", "A", "A", "B"],
                      "item_id": ["I1", "I2", "I3", "I4"]})
print("F Joined rows =", len(items), "| distinct orders =", items["order_id"].nunique())
cases = [(72, 80), (54, 60), (30, 50)]
print("G RiverTown rates =", ", ".join(pct(a / e) for a, e in cases))
print("H MetroMeal latest rate =", pct(0.784), "| gap =", f"{gap(0.784, 0.85):.1f}", "pp")
print("I GreenBlock completion =", pct(273 / 300),
      "| gap =", f"{gap(273 / 300, 0.95):.1f}", "pp")
print("J MetroMeal reconcile 392 + 108 - 500 =", 392 + 108 - 500)

# ---- Part K: source, key, grain and eligibility ----
df = pd.read_csv("orders_story.csv")
print("\nK Rows =", len(df), "| distinct order_id =", df["order_id"].nunique(),
      "| eligible =", int(df["eligible_kpi_flag"].sum()))
target, guardrail = df["target_rate"].max(), df["complaint_guardrail"].max()

# ---- Part L: weekly KPI table, count-first aggregate, latest late checks ----
elig = df[df["eligible_kpi_flag"] == 1]
weekly = elig.groupby("week_start").agg(eligible=("order_id", "size"),
                                       on_time=("on_time_kpi_flag", "sum"),
                                       complaints=("complaint_flag", "sum"))
weekly["rate"] = weekly["on_time"] / weekly["eligible"]
weekly["gap_pp"] = gap(weekly["rate"], target)
weekly["complaint_rate"] = weekly["complaints"] / weekly["eligible"]
print("\nL Weekly KPI table")
```

```

formats = {"rate": pct, "complaint_rate": pct, "gap_pp": "{:.1f}".format}
print(weekly.to_string(formatters=formats))
total = weekly[["eligible", "on_time", "complaints"]].sum()
overall = total["on_time"] / total["eligible"]
print("Four-week aggregate:", int(total["on_time"]), "/", int(total["eligible"]),
      "=", pct(overall, 2), "| gap =", f"{gap(overall, target):.2f}", "pp",
      "| complaints =", int(total["complaints"]),
      "=", pct(total["complaints"] / total["eligible"], 2))
latest = weekly.index.max()
is_late = (elig["week_start"] == latest) & (elig["late_minutes"] > 0)
late = elig.loc[is_late, "late_minutes"]
row = weekly.loc[latest]
print("Latest week", latest, ": late =", len(late), "| mean =", round(late.mean(), 1),
      "| median =", late.median(),
      "| 90th percentile =", late.quantile(0.9),          # same as PERCENTILE.INC
      "| reconcile =", int(row["eligible"] - row["on_time"] - len(late)))
ok = row["rate"] < target and row["complaint_rate"] <= guardrail
print("Decision check:", "Investigate, guardrail within limit" if ok
      else "Review KPI and guardrail")

# ---- Part M: direct labels for the chart ----
labels = [f"{pct(r)} (n = {n})" for r, n in zip(weekly["rate"], weekly["eligible"])]
print("\nM Labels:", ", ".join(labels))

# ---- Part M: the repaired chart, saved as a PNG ----
fig, ax = plt.subplots(figsize=(7, 4))
bars = ax.bar(weekly.index, weekly["rate"], color="#9aa9c9", edgecolor="black")
bars[-1].set_hatch("/")          # non-color cue on the latest week
ax.axhline(target, color="black", linestyle="--", linewidth=1.5,
            label=f"Target {target:.1%}")
for bar, r, label in zip(bars, weekly["rate"], labels):
    ax.text(bar.get_x() + bar.get_width() / 2, r + 0.02, label, ha="center", fontsize=9)
ax.set_ylim(0, 1)              # zero baseline for bars
ax.yaxis.set_major_formatter(PercentFormatter(1.0))
ax.set_ylabel("On-time rate")
ax.set_title("Latest observed weekly on-time rate was below the 85.0% target")
ax.legend(loc="lower left")
fig.tight_layout()
fig.savefig("S02P_repaired_chart.png", dpi=150)
print("Saved S02P_repaired_chart.png")

```

The same checks in base R, plus the repaired chart saved as `S02P_repaired_chart_r.png` . `quantile(type = 7)` matches Excel's `PERCENTILE.INC` , so the 90th percentile should print 31.

```
# S02P Parts A-N in R (optional, not graded). Base R only.
# Put this file next to orders_story.csv.

pct <- function(x, d = 1) sprintf(paste0("%.", d, "f%%"), 100 * x)
gap <- function(rate, target) (rate - target) * 100      # percentage points
g1 <- function(rate, target) sprintf("%.1f", gap(rate, target))

# ---- Parts A-J: the five practice cases (typed numbers) ----
cat("A HarborCare no-show rate =", pct(46 / 400), "| complaint rate =", pct(12 / 400), "\n")
cat("B MetroMeal on-time rate =", pct(392 / 500), "| gap =", g1(392 / 500, 0.85),
    "pp | complaint rate =", pct(14 / 500), "\n")
cat("C NorthLine resolution =", pct(684 / 760), "| gap =", g1(684 / 760, 0.92), "pp\n")
on_time_w <- c(420, 438, 425, 392); eligible_w <- rep(500, 4)
cat("D MetroMeal four-week rate =", pct(sum(on_time_w) / sum(eligible_w), 2), "\n")
cat("E BrightCart joined rows =", 310, "item rows for", 120, "orders\n")
items <- data.frame(order_id = c("A", "A", "A", "B"), item_id = c("I1", "I2", "I3", "I4"))
cat("F Joined rows =", nrow(items),
    "| distinct orders =", length(unique(items$order_id)), "\n")
cat("G RiverTown rates =", pct(c(72, 54, 30) / c(80, 60, 50)), "\n")
cat("H MetroMeal latest rate =", pct(0.784), "| gap =", g1(0.784, 0.85), "pp\n")
cat("I GreenBlock completion =", pct(273 / 300), "| gap =", g1(273 / 300, 0.95), "pp\n")
cat("J MetroMeal reconcile 392 + 108 - 500 =", 392 + 108 - 500, "\n")

# ---- Part K: source, key, grain and eligibility ----
df <- read.csv("orders_story.csv", stringsAsFactors = FALSE)
cat("\nK Rows =", nrow(df), "| distinct order_id =", length(unique(df$order_id)),
    "| eligible =", sum(df$eligible_kpi_flag), "\n")
target <- max(df$target_rate); guardrail <- max(df$complaint_guardrail)

# ---- Part L: weekly KPI table, aggregate and latest late checks ----
elig <- df[df$eligible_kpi_flag == 1, ]
weekly <- aggregate(cbind(on_time = on_time_kpi_flag, complaints = complaint_flag,
    eligible = eligible_kpi_flag) ~ week_start,
    data = elig, FUN = sum)
weekly$rate <- weekly$on_time / weekly$eligible
weekly$gap_pp <- round(gap(weekly$rate, target), 1)
weekly$complaint_rate <- weekly$complaints / weekly$eligible
print(transform(weekly, rate = pct(rate), complaint_rate = pct(complaint_rate)))
overall <- sum(weekly$on_time) / sum(weekly$eligible)
cat("Four-week aggregate:", sum(weekly$on_time), "/", sum(weekly$eligible),
    "=", pct(overall, 2), "| gap =", sprintf("%.2f", gap(overall, target)), "pp",
    "| complaints =", sum(weekly$complaints),
    "=", pct(sum(weekly$complaints) / sum(weekly$eligible), 2), "\n")
latest <- max(weekly$week_start)      # ISO dates sort correctly as text
late <- elig$late_minutes[elig$week_start == latest & elig$late_minutes > 0]
last <- weekly[weekly$week_start == latest, ]
p90 <- quantile(late, 0.9, type = 7, names = FALSE)  # type 7 = PERCENTILE.INC
cat("Latest week", latest, ": late =", length(late), "| mean =", round(mean(late), 1),
    "| median =", median(late), "| 90th percentile =", p90,
    "| reconcile =", last$eligible - last$on_time - length(late), "\n")
ok <- last$rate < target & last$complaint_rate <= guardrail
cat("Decision check:", ifelse(ok, "Investigate, guardrail within limit",
```

```

"Review KPI and guardrail"), "\n")

# ---- Part M: direct labels for the chart ----
labels <- paste0(pct(weekly$rate), " (n = ", weekly$eligible, ")")
cat("\nM Labels:", labels, sep = "\n ")

# ---- Part M: the repaired chart, saved as a PNG ----
png("S02P_repaired_chart_r.png", width = 900, height = 520)
mids <- barplot(weekly$rate, names.arg = weekly$week_start,
               ylim = c(0, 1),                               # zero baseline
               col = "#9aa9c9", density = c(-1, -1, -1, 20), # hatch the latest week
               ylab = "On-time rate",
               main = "Latest observed weekly on-time rate was below the 85.0% target")
abline(h = target, lty = 2, lwd = 2)
text(mids, weekly$rate + 0.04, labels)
legend("bottomleft", legend = paste("Target", pct(target)), lty = 2, lwd = 2, bg = "white")
dev.off()
cat("\nSaved S02P_repaired_chart_r.png\n")

```

## Topic 2 · Weekday and weekend on-time rates, week by week Extend

Same question in Excel → SPSS → Python → R.

### Excel Parts L

Click inside the **Orders** table and choose **Insert > PivotTable > New Worksheet**. Put **week\_start** in Rows and **weekend\_flag** in Columns. If Excel groups the dates into months, right-click a date and choose **Ungroup**. Put **on\_time\_kpi\_flag** in Values three times: **Sum** (on-time count), **Count** (denominator) and **Average** (the average of a 0/1 flag is the on-time rate; format it as a percentage). The latest week should show 320 of 358 (89.4%) on weekdays and 72 of 142 (50.7%) on weekends. Write one bounded sentence about it.

### SPSS Parts L

The crosstab shows on-time counts and row percentages for weekdays and weekends in each week. Compare the latest week with the earlier three.

New code: paste it at the end of **S02P\_SPSS.sps**, make sure the setup has run this session, then select only the pasted lines and choose **Run > Selection**.

```

* Extension: on-time deliveries by weekend flag within each week.
* Run the import block first so OrdersStory is open.
DATASET ACTIVATE OrdersStory.
TEMPORARY.
SELECT IF (eligible_kpi_flag=1).
CROSSTABS
  /TABLES=weekend_flag BY on_time_kpi_flag BY week_start
  /CELLS=COUNT ROW.

```

## Python Parts L

Split each week's on-time rate into weekday and weekend deliveries, with every denominator. Then ask: does a lower weekend rate show that staffing *caused* it? What else differs on weekends?

```
# Extension: is the latest dip concentrated on weekends? (descriptive only)
import pandas as pd

df = pd.read_csv("orders_story.csv")
elig = df[df["eligible_kpi_flag"] == 1]
table = elig.pivot_table(index="week_start", columns="weekend_flag",
                          values="on_time_kpi_flag", aggfunc=["sum", "count"])

for week in table.index:
    parts = []
    for flag, name in [(0, "weekday"), (1, "weekend")]:
        on_time, n = table.loc[week, ("sum", flag)], table.loc[week, ("count", flag)]
        parts.append(f"{name} {on_time}/{n} = {on_time / n:.1f}%")
    print(week, " | ".join(parts))
print("Question: does this show that weekend staffing *caused* the dip?")
print("What else differs on weekends?")
```

## R Parts L

The same split in base R with `tapply()`. The latest week should show weekday 320/358 = 89.4% and weekend 72/142 = 50.7%.

```
# Extension: is the latest dip concentrated on weekends? (descriptive only)
# Base R only.
df <- read.csv("orders_story.csv", stringsAsFactors = FALSE)
elig <- df[df$eligible_kpi_flag == 1, ]
groups <- list(elig$week_start, elig$weekend_flag)
on_time <- tapply(elig$on_time_kpi_flag, groups, sum)      # rows = weeks, columns = 0/1
n <- tapply(elig$on_time_kpi_flag, groups, length)
rate <- 100 * on_time / n
for (week in rownames(on_time)) {
    cat(week, sprintf("weekday %d/%d = %.1f%% | weekend %d/%d = %.1f%%\n",
                      on_time[week, "0"], n[week, "0"], rate[week, "0"],
                      on_time[week, "1"], n[week, "1"], rate[week, "1"])))
}
cat("Question: does this show that weekend staffing *caused* the dip?\n")
cat("What else differs on weekends?\n")
```

## Topic 3 · The latest week by region Extend

Same question in Excel → SPSS → Python → R.

## Excel Parts L

Insert another PivotTable from the `Orders` table. Put `week_start` in **Filters** and pick 2026-08-24. Put `region` in Rows and `on_time_kpi_flag` in Values three times: **Sum**, **Count** and **Average** (as a percentage). The four counts should add to 500. Are the differences between regions large compared with the size of each group?

## SPSS Parts L

The crosstab shows on-time counts and row percentages for each region in the latest week. The row totals are the denominators and should add to 500.

New code: paste it at the end of `S02P_SPSS.sps`, make sure the setup has run this session, then select only the pasted lines and choose **Run > Selection**.

```
* Extension: latest-week on-time deliveries by region, with each denominator.
* Run the import block first so OrdersStory is open.
DATASET ACTIVATE OrdersStory.
TEMPORARY.
SELECT IF (week_start=DATE.DMY(24,8,2026) AND eligible_kpi_flag=1).
CROSSTABS
  /TABLES=region BY on_time_kpi_flag
  /CELLS=COUNT ROW.
```

## Python Parts L

Break the latest week into the four regions with their denominators. Are the differences large compared with the size of each group?

```
# Extension: latest-week on-time rate by region, with each denominator.
import pandas as pd

df = pd.read_csv("orders_story.csv")
is_latest = df["week_start"] == df["week_start"].max()
latest = df[(df["eligible_kpi_flag"] == 1) & is_latest]
by_region = latest.groupby("region").agg(on_time=("on_time_kpi_flag", "sum"),
                                         n=("order_id", "size"))
by_region["rate"] = (by_region["on_time"] / by_region["n"]).map("{:.1%}".format)
print(by_region.to_string())
print(f"Check: the four n values add to {by_region['n'].sum()} (should be {len(latest)}).")
print("Question: is any region far from the others, or are the gaps small next to n?")
```

## R Parts L

The same breakdown in base R with `aggregate()`. The four n values should add to 500.

```
# Extension: latest-week on-time rate by region, with each denominator.
df <- read.csv("orders_story.csv", stringsAsFactors = FALSE)
latest <- df[df$eligible_kpi_flag == 1 & df$week_start == max(df$week_start), ]
by_region <- aggregate(cbind(on_time = on_time_kpi_flag, n = eligible_kpi_flag) ~ region,
                      data = latest, FUN = sum)
by_region$rate <- sprintf("%.1f%%", 100 * by_region$on_time / by_region$n)
print(by_region)
cat("Check: the four n values should add to", nrow(latest), "\n")
cat("Question: is any region far from the others, or are the gaps small next to n?\n")
```

## Topic 4 · How late are the late deliveries, week by week? Extend

Same question in Excel → SPSS → Python → R.

### Excel Parts L

In a *copy* of your workbook, add a new sheet. In A2 type `=DATE(2026,8,3)`, in A3 type `=A2+7` and fill down to A5. In B2 type `=COUNTIFS(Orders[week_start],A2,Orders[late_minutes],">0")`. In C2 type `=AVERAGEIFS(Orders[late_minutes],Orders[week_start],A2,Orders[late_minutes],">0")`. In D2 type `=MEDIAN(FILTER(Orders[late_minutes],(Orders[week_start]=A2)*(Orders[late_minutes]>0)))`, and in E2 type `=PERCENTILE.INC(FILTER(Orders[late_minutes],(Orders[week_start]=A2)*(Orders[late_minutes]>0)),0.9)`. Fill row 2 down to row 5. Every row in this file is eligible (Part K), so no eligibility test is needed here. The latest week should match `KPI_Build` row 18: 108, 12.4, 7 and 31.

### SPSS Parts L

**EXAMINE** describes the late deliveries in each week, with medians, 90th percentiles and boxplots. SPSS uses a different percentile rule from `PERCENTILE.INC`, so small differences from Excel are expected.

New code: paste it at the end of `S02P_SPSS.sps`, make sure the setup has run this session, then select only the pasted lines and choose **Run > Selection**.

```
* Extension: late minutes week by week, for late deliveries only.
* Run the import block first so OrdersStory is open.
DATASET ACTIVATE OrdersStory.
TEMPORARY.
SELECT IF (eligible_kpi_flag=1 AND late_minutes>0).
EXAMINE VARIABLES=late_minutes BY week_start
  /PLOT=BOXPLOT
  /STATISTICS=DESCRIPTIVES
  /PERCENTILES(50,90)
  /NOTOTAL.
```

## Python Parts L

Summarize late minutes week by week and save a boxplot as a PNG. The latest week should show  $n = 108$ , mean 12.4, median 7 and 90th percentile 31.

```
# Extension: mean, median and 90th percentile of late minutes for every week.
import matplotlib
matplotlib.use("Agg")                # save to file; no window needed
import matplotlib.pyplot as plt
import pandas as pd

df = pd.read_csv("orders_story.csv")
late = df[(df["eligible_kpi_flag"] == 1) & (df["late_minutes"] > 0)]
summary = late.groupby("week_start")["late_minutes"].agg(
    n="size", mean="mean", median="median",
    p90=lambda x: x.quantile(0.9))    # same as PERCENTILE.INC
print(summary.round(1).to_string())

weeks = sorted(late["week_start"].unique())
fig, ax = plt.subplots(figsize=(7, 4))
ax.boxplot([late.loc[late["week_start"] == w, "late_minutes"] for w in weeks])
ax.set_xticks(range(1, len(weeks) + 1), weeks)
ax.set_ylabel("Minutes late"); ax.set_title("Late deliveries by week")
fig.tight_layout()
fig.savefig("late_minutes_boxplot.png", dpi=150)
print("Saved late_minutes_boxplot.png")
print("Question: in which week is the mean furthest above the median?")
print("What does that suggest?")
```

## R Parts L

The same summary in base R, with a boxplot. In which week is the mean furthest above the median?

```
# Extension: mean, median and 90th percentile of late minutes for every week.
df <- read.csv("orders_story.csv", stringsAsFactors = FALSE)
late <- df[df$eligible_kpi_flag == 1 & df$late_minutes > 0, ]
summ <- function(x) {
  c(n = length(x), mean = round(mean(x), 1), median = median(x),
    p90 = quantile(x, 0.9, type = 7, names = FALSE))    # type 7 = PERCENTILE.INC
}
out <- do.call(rbind, lapply(split(late$late_minutes, late$week_start), summ))
print(out)
boxplot(late_minutes ~ week_start, data = late, ylab = "Minutes late",
  main = "Late deliveries by week")
cat("Question: in which week is the mean furthest above the median?\n")
cat("What does that suggest?\n")
```

## Topic 5 · An average of rates versus the rate from counts Extend

Same question in Excel → SPSS → Python → R.

### Excel Parts D

In a *copy* of your workbook, go to `Slide_Practice` and make the latest week twice as big: change `B24` from 392 to 784 and `B28` from 500 to 1000 (its rate is still 78.4%). In an empty cell type `=AVERAGE(B21/B25,B22/B26,B23/B27,B24/B28)`. It still shows 83.75%, but `B34` drops to 82.68%. Which one counts every delivery once?

### SPSS Parts D

This block types its own numbers, like Hand-off Block H2, and opens a new small dataset. **DESCRIPTIVES** gives the mean of the four rates, 83.75%. **AGGREGATE** sums the counts first and gives 82.68%.

New code: paste it at the end of `S02P_SPSS.sps`, make sure the setup has run this session, then select only the pasted lines and choose **Run > Selection**.

```
* Extension: an average of rates can hide unequal week sizes.
* This block types its own numbers, like Hand-off Block H2.
* The latest week is made twice as big. Its rate is still 78.4%.
NEW FILE.
DATA LIST FREE / week (A10) ontime eligible.
BEGIN DATA
2026-08-03 420 500
2026-08-10 438 500
2026-08-17 425 500
2026-08-24 784 1000
END DATA.
COMPUTE rate=ontime/eligible.
FORMATS rate (PCT6.2).
LIST.

* Route 1: average the four rates. Each week counts once.
DESCRIPTIVES VARIABLES=rate
  /STATISTICS=MEAN.

* Route 2: sum the counts, then divide. Each delivery counts once.
AGGREGATE
  /OUTFILE=*
  /BREAK=
  /ontime_total=SUM(ontime)
  /eligible_total=SUM(eligible).
COMPUTE overall_rate=ontime_total/eligible_total.
FORMATS overall_rate (PCT6.2).
LIST.
```

## Python Parts D

Types the same four weeks with the latest week doubled. You should see 83.75% for the average of rates and  $2067 / 2500 = 82.68\%$  from the counts.

```
# Extension: watch an average of rates go wrong when weeks have different sizes.
# The latest week is made twice as big (784 of 1000). Its rate is still 78.4%.
on_time = [420, 438, 425, 784]
eligible = [500, 500, 500, 1000]

rates = [o / e for o, e in zip(on_time, eligible)]
print("Weekly rates:", ", ".join(f"{r:.1%}" for r in rates))

average_of_rates = sum(rates) / len(rates)          # each week counts once
rate_from_counts = sum(on_time) / sum(eligible)    # each delivery counts once
print(f"Average of the four rates : {average_of_rates:.2%}")
print(f"Sum counts, then divide : {sum(on_time)} / {sum(eligible)} "
      f"= {rate_from_counts:.2%}")
print("Question: which number counts every delivery once?")
```

## R Parts D

The same two routes in base R. You should see 83.75% and 82.68%.

```
# Extension: watch an average of rates go wrong when weeks have different sizes.
# Base R only. The latest week is made twice as big (784 of 1000).
# Its rate is still 78.4%.
on_time <- c(420, 438, 425, 784)
eligible <- c(500, 500, 500, 1000)

rates <- on_time / eligible
cat("Weekly rates:", sprintf("%.1f%%", 100 * rates), "\n")
# Route 1: each week counts once.
cat(sprintf("Average of the four rates : %.2f%%\n", 100 * mean(rates)))
# Route 2: each delivery counts once.
cat(sprintf("Sum counts, then divide : %d / %d = %.2f%%\n",
            sum(on_time), sum(eligible), 100 * sum(on_time) / sum(eligible)))
cat("Question: which number counts every delivery once?\n")
```